This is CS50.

0-SCRATCH

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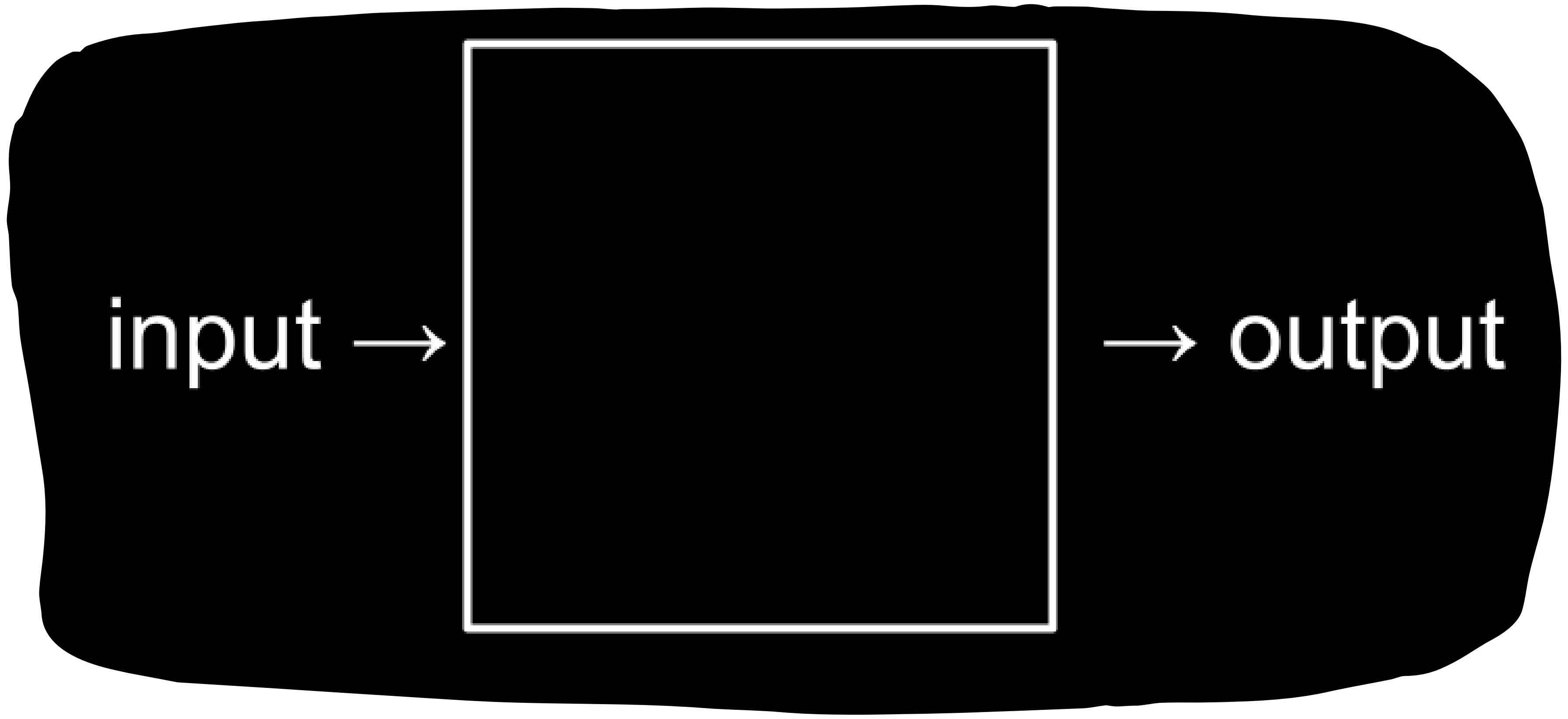
Week 0 : SCRATCH

📌 Outline :

- WELCOME!
- COMPUTER SCIENCE
- REPRESENTATION
- ASCII
- SCRATCH
- HELLO WORLD
- HELLO, YOU
- MEOW AND ABSTRACTION
- CONDITIONALS

computer science

- Essentially, computer programming is about taking some input and creating some output - thus solving a problem. What happens in between the input and output, what we could call a black box, is the focus of this course.



برمجة الكمبيوتر هى عبارة عن اخذ مدخلات من المستخدم وانشاء بعض المخرجات ليها وده اللى يسمى ب حل المشكلات (PROBLEM SOLVING) : العملية اللى بتحصل بين المدخلات والمخرجات وبنسميه BLACK BOX .
ده تحديدا محور الكورس واللى هناقشه فى الكورس كله .

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طب أنا دلوقتي كمبرمج عايز ابعت
للكمبيوتر المدخلات او ال INPUTS
بتاعتى هعمل كده ازاى او هتواصل مع
الكمبيوتر ازاى ؟

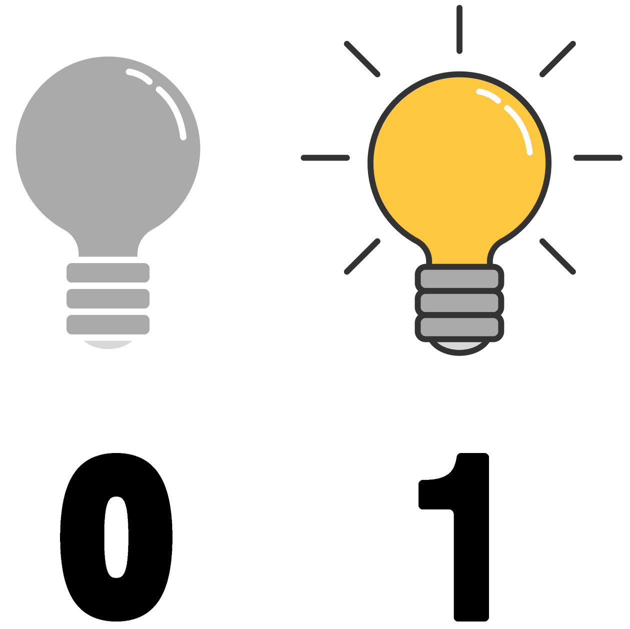
اى اللغة الى بيفهمها
الكومبيوتر ؟

Representation

- we may need to take attendance for a class. We could use a system called unary to count, one finger at a time.

- Computers only speak in terms of zeros and ones. Zeros represent off. Ones represent on. Computers are millions, and perhaps billions, of transistors that are being turned on and off.

- If you imagine using a light bulb, a single bulb can only count from zero to one.



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Binary:

- A binary system is a number system that uses only two symbols or digits, typically 0 and 1, to represent numbers. It is also known as the base-2 number system.

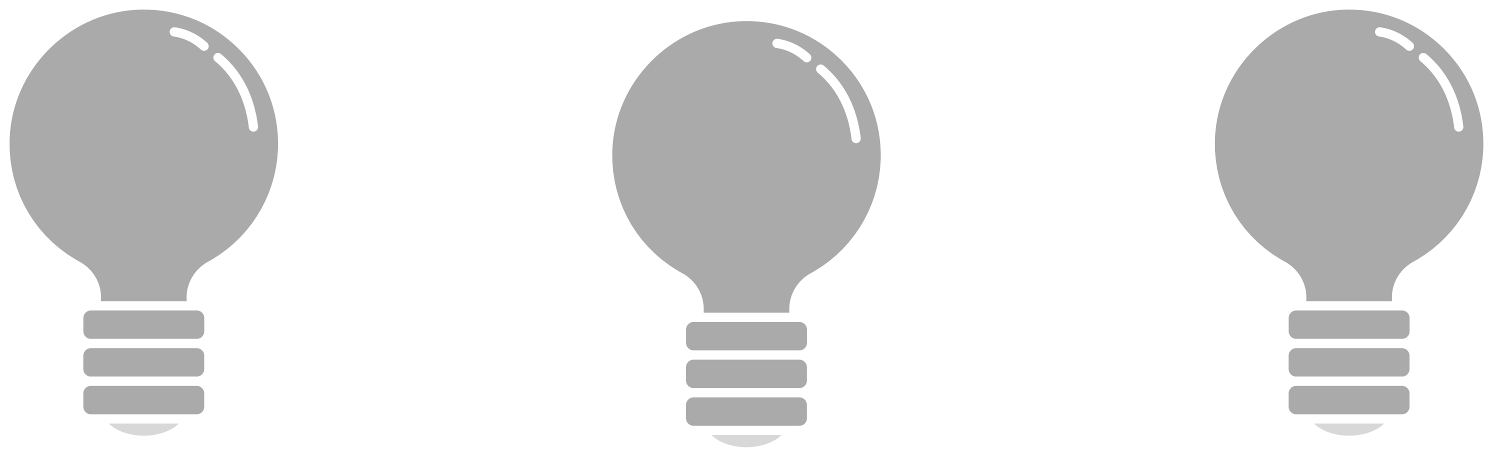
- Computers today count using binary system. It’s from the term binary digit that we get a familiar term called bit.

- A bit is a zero or one : on or off.

1 0

Binary

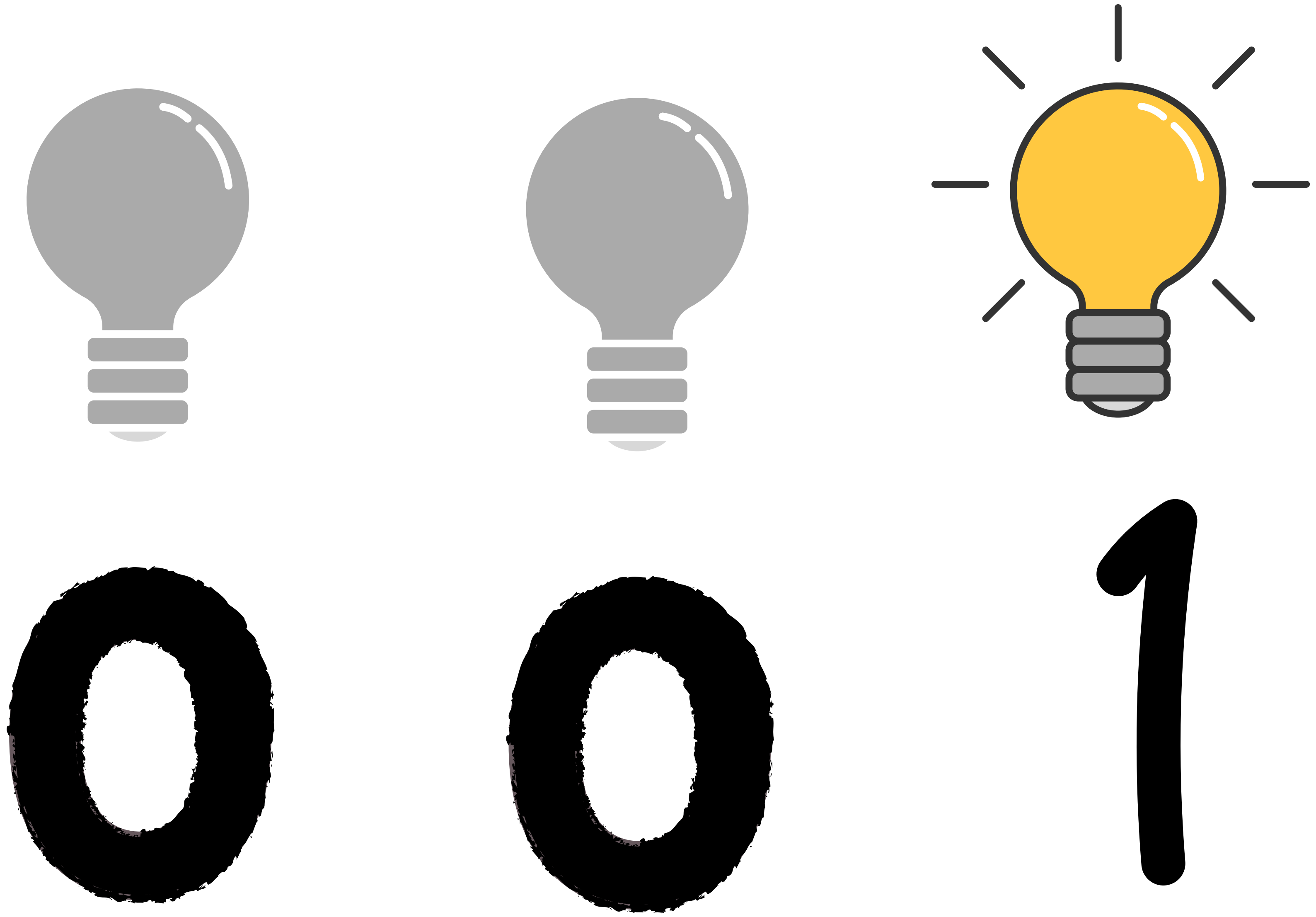
However, if you were to have three light bulbs, there are more options open to you!
Similarly, the following would represent zero:



0 0 0

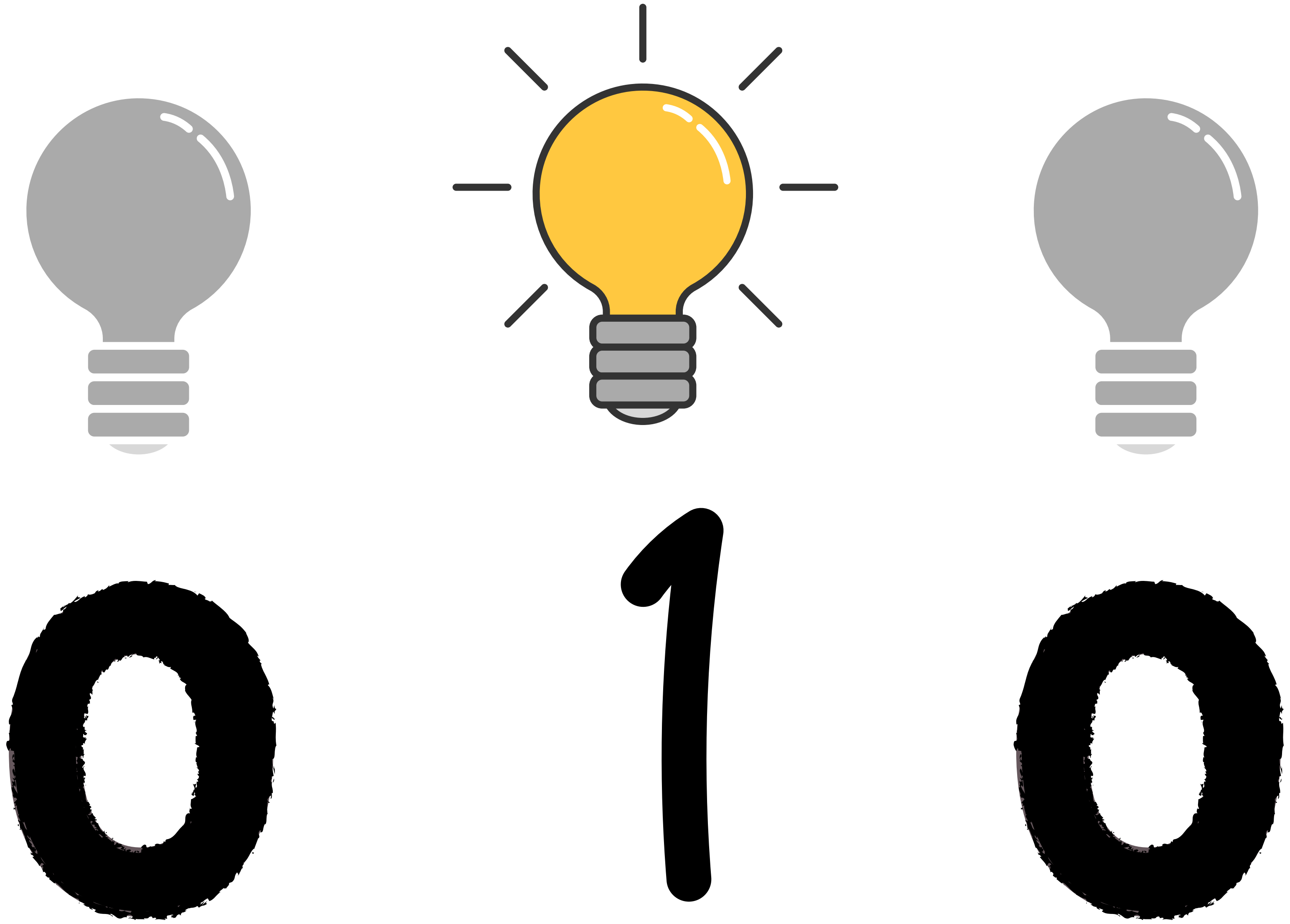
Binary

The following would represent one:



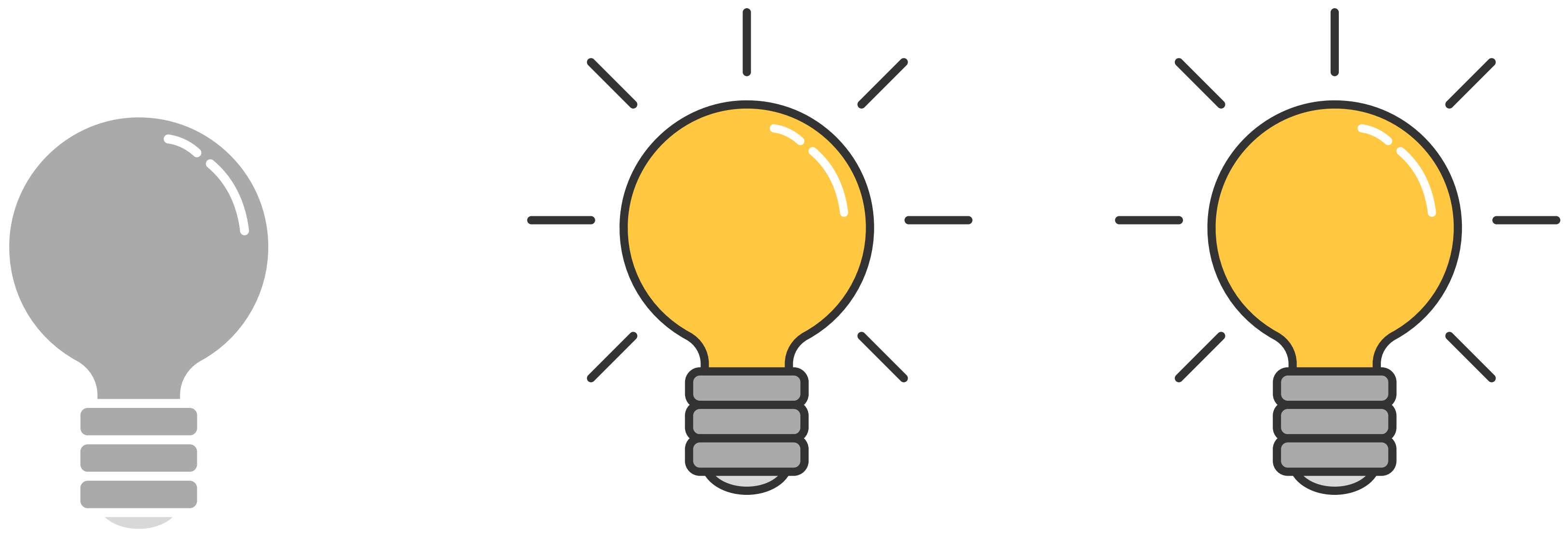
Binary

The following would represent two:



Binary

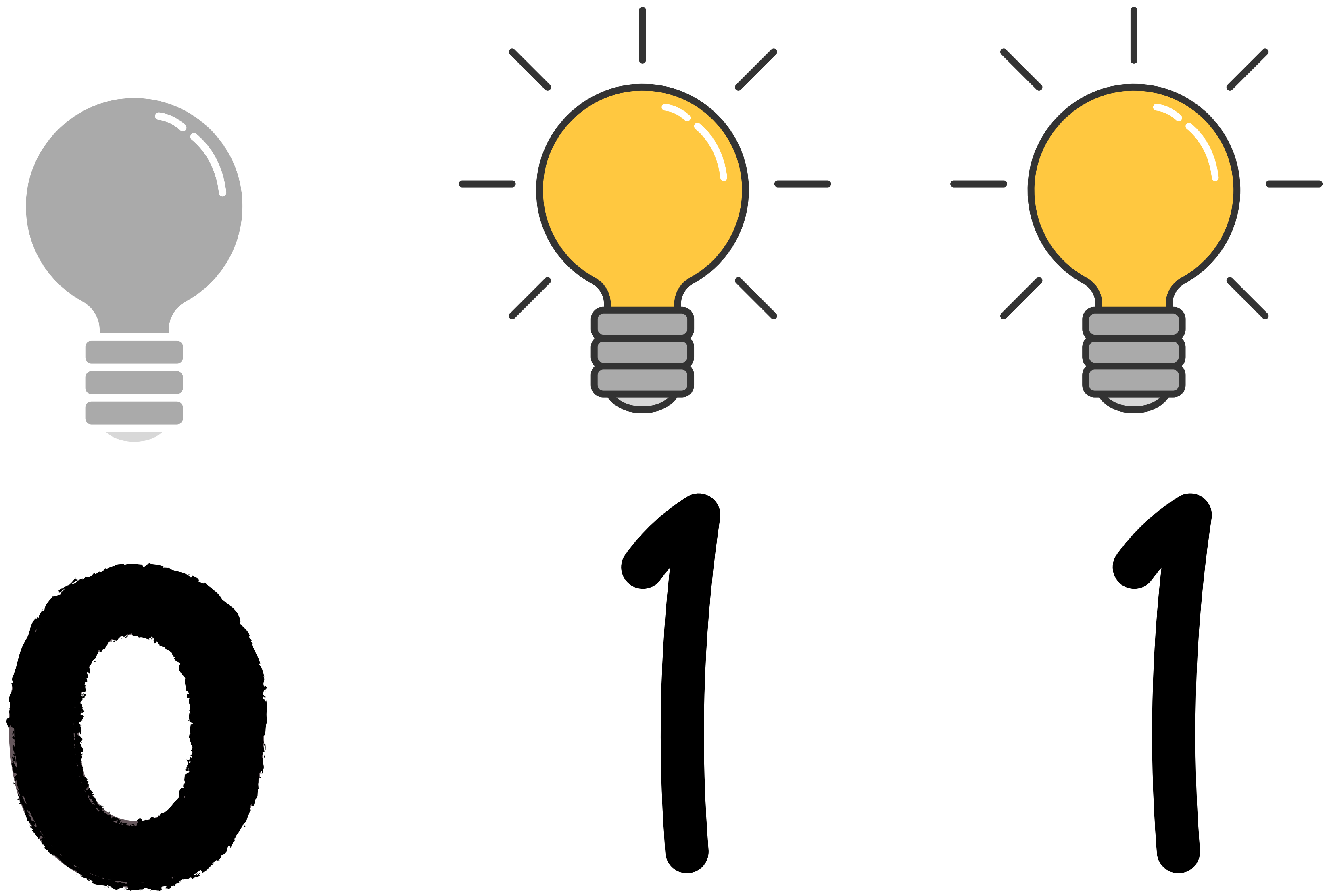
The following would represent three:



0 1 1

Binary

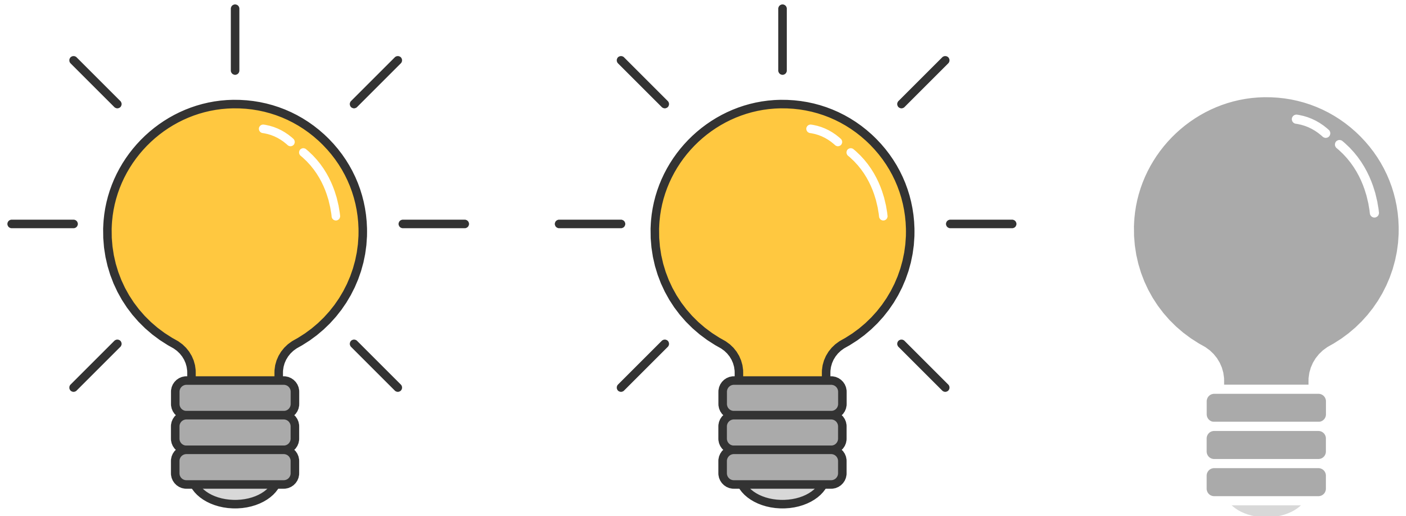
The following would represent three:



Binary

the following would represent six:

2 1 0



2^2 2^1 2^0

4 2 1

110

6 = 1*4 + 1*2 + 0*1

Binary

The following would represent what?:

1 1 1

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Byte

- Therefore, you could say that it would require three bits (the four's place, the two's place, and the one's place) to represent a number as high as seven.
- Computers generally use eight bits to represent a number.

- byte = 8 bits .
- For example,

.

00000101

is the number 5 in binary.

ASCII

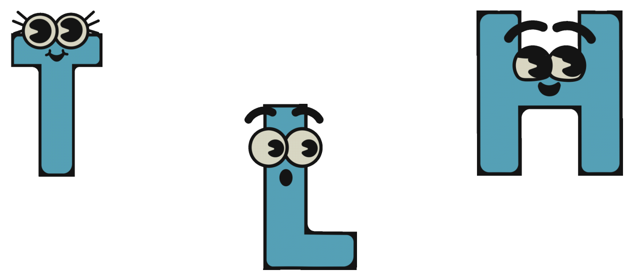
- JUST AS NUMBERS ARE BINARY PATTERNS OF ONES AND ZEROS, LETTERS ARE REPRESENTED USING ONES AND ZEROS TOO!
- SINCE THERE IS AN OVERLAP BETWEEN THE ONES AND ZEROS THAT REPRESENT NUMBERS AND LETTERS, THE ASCII STANDARD WAS CREATED TO MAP SPECIFIC LETTERS TO SPECIFIC NUMBERS.
- FOR EXAMPLE,

THE LETTER A WAS DECIDED TO MAP TO THE NUMBER 65.

A

65

ASCII



- IF YOU RECEIVED A TEXT MESSAGE, THE BINARY UNDER THAT MESSAGE MIGHT REPRESENT THE NUMBERS 72, 73, AND 33. MAPPING THESE OUT TO ASCII, YOUR MESSAGE WOULD LOOK AS FOLLOWS

01001000	**01001001**	**00100001**
72	**73**	**33**
H	I	!

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ASCII

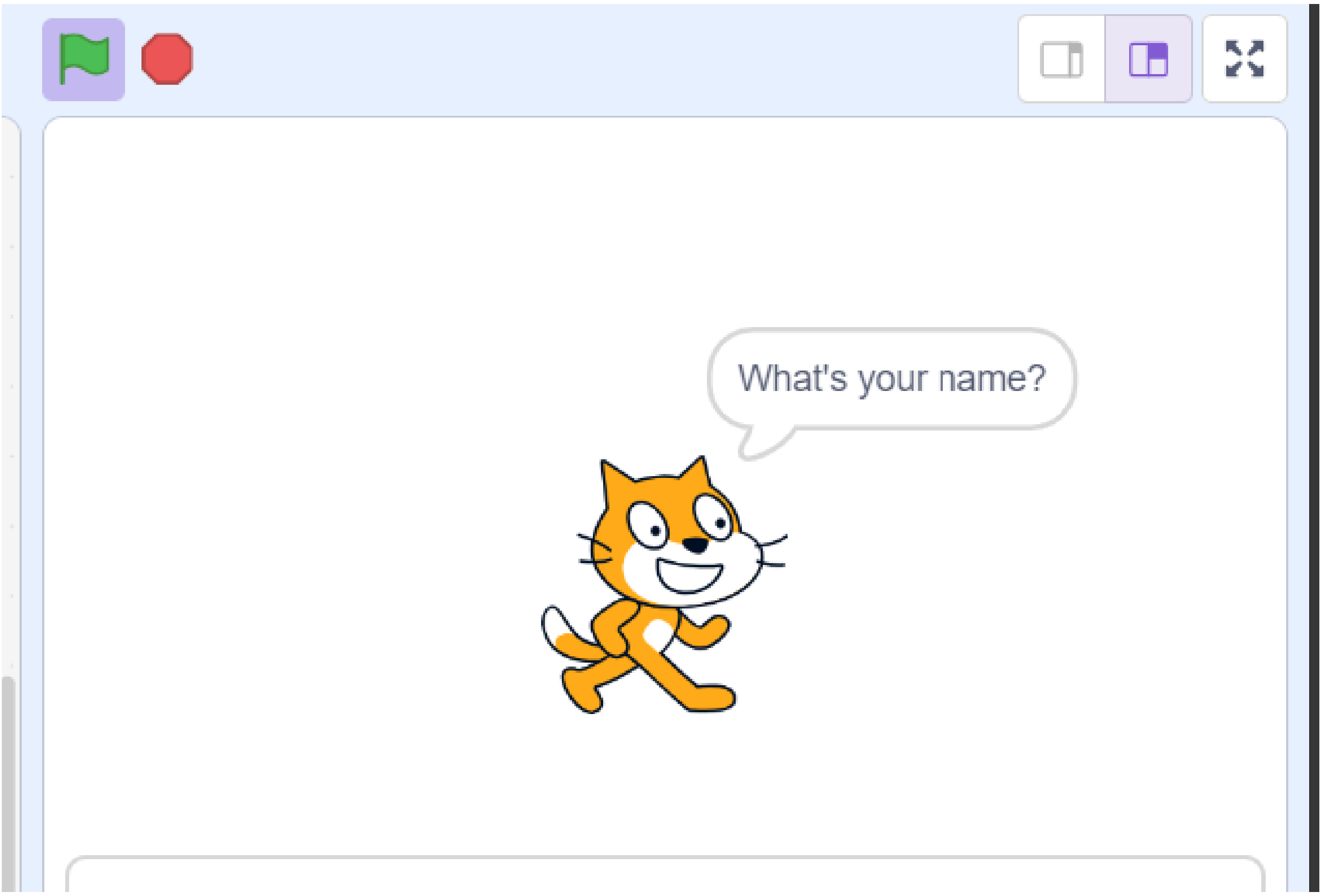
HERE IS AN EXPANDED MAP OF ASCII VALUES:

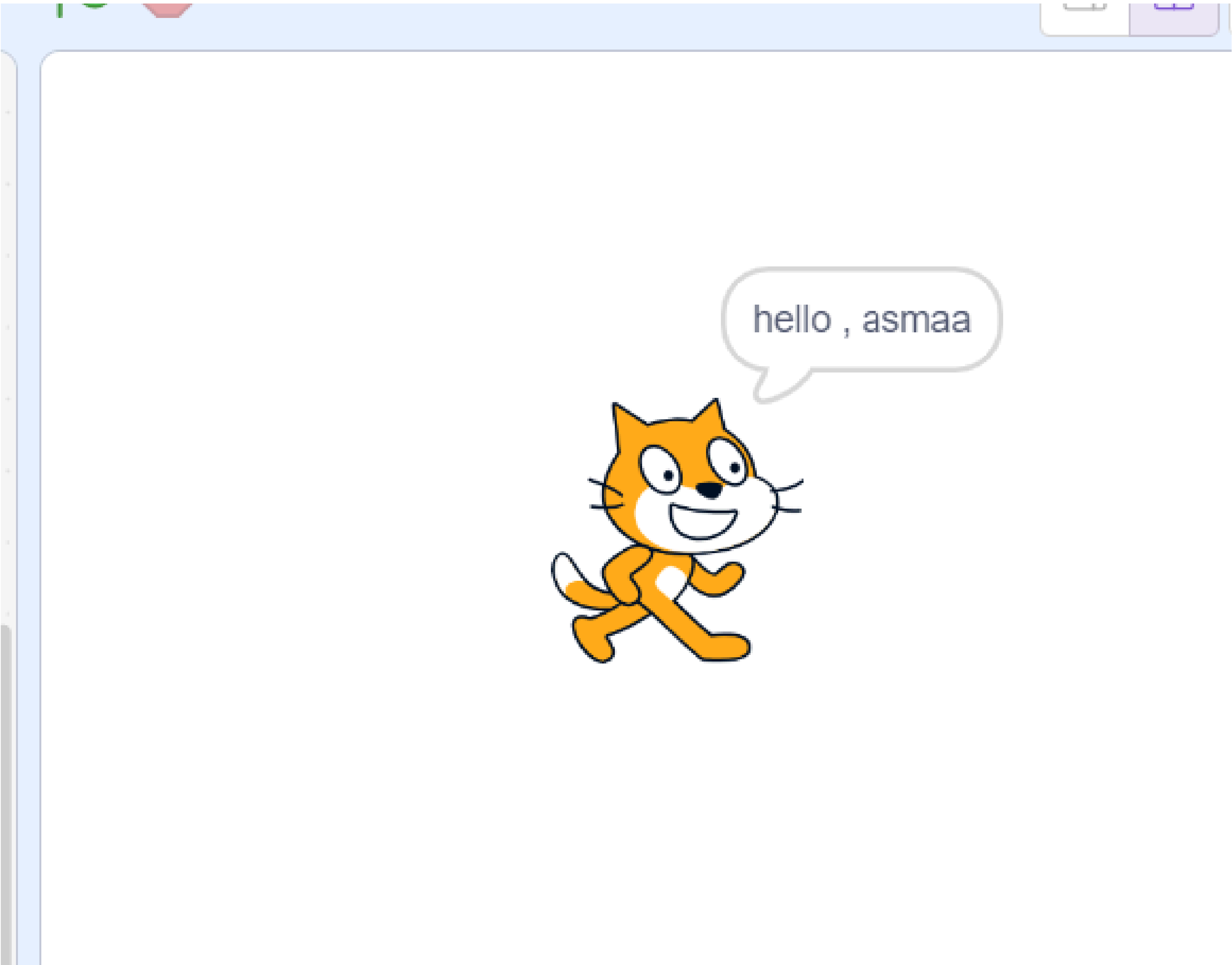
0	NUL	16	DLE	32	SP	48	0	64	@	80	P	96	`	112	p
1	SOH	17	DC1	33	!	49	1	65	A	81	Q	97	a	113	q
2	STX	18	DC2	34	"	50	2	66	B	82	R	98	b	114	r
3	ETX	19	DC3	35	#	51	3	67	C	83	S	99	c	115	s
4	EOT	20	DC4	36	$	52	4	68	D	84	T	100	d	116	t
5	ENQ	21	NAK	37	%	53	5	69	E	85	U	101	e	117	u
6	ACK	22	SYN	38	&	54	6	70	F	86	V	102	f	118	v
7	BEL	23	ETB	39	'	55	7	71	G	87	W	103	g	119	w
8	BS	24	CAN	40	(	56	8	72	H	88	X	104	h	120	x
9	HT	25	EM	41	)	57	9	73	I	89	Y	105	i	121	y
10	LF	26	SUB	42	*	58	:	74	J	90	Z	106	j	122	z
11	VT	27	ESC	43	+	59	;	75	K	91	[	107	k	123	{
12	FF	28	FS	44	,	60	<	76	L	92	\	108	l	124	|
13	CR	29	GS	45	-	61	=	77	M	93	]	109	m	125	}
14	SO	30	RS	46	.	62	>	78	N	94	^	110	n	126	~
15	SI	31	US	47	/	63	?	79	O	95	_	111	o	127	DEL

SCRATCH

- loops
- functions
- conditionals

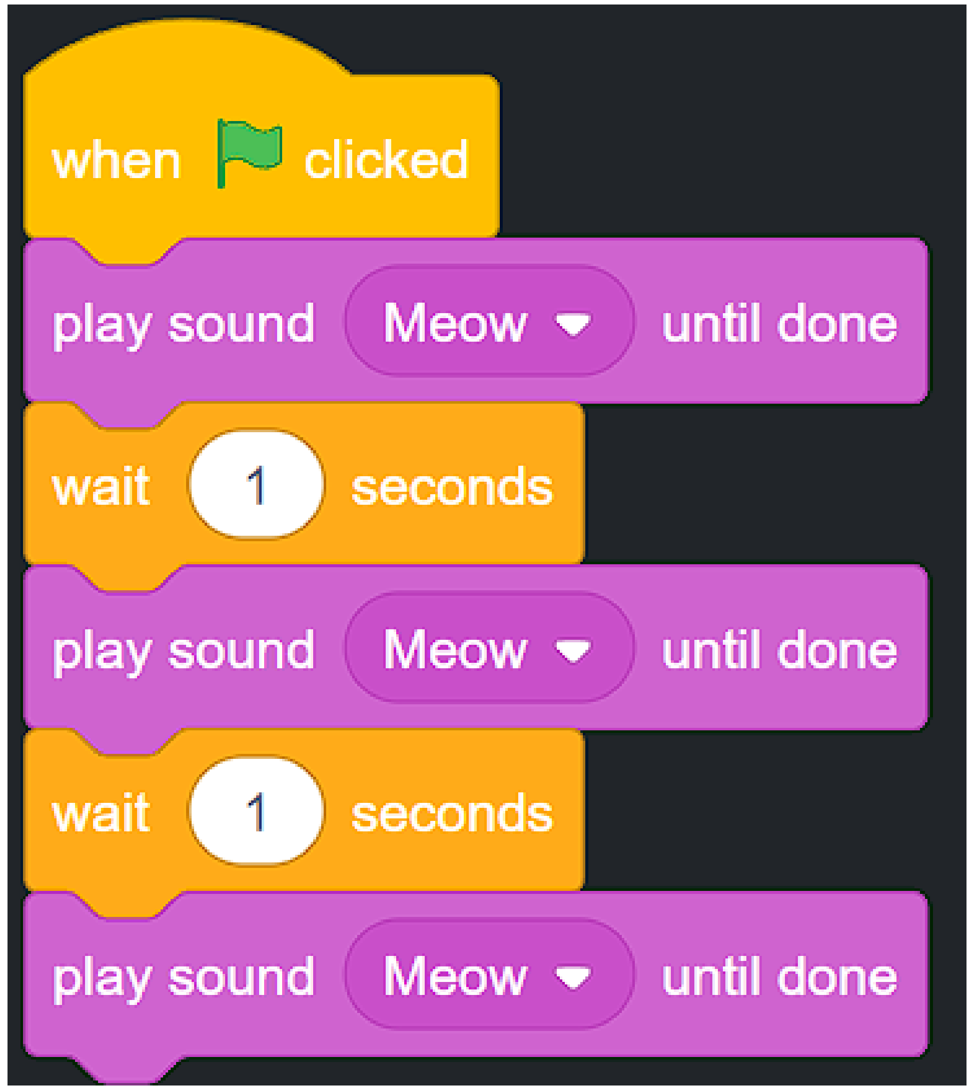

when clicked
ask What's your name? and wait
say join hello, answer

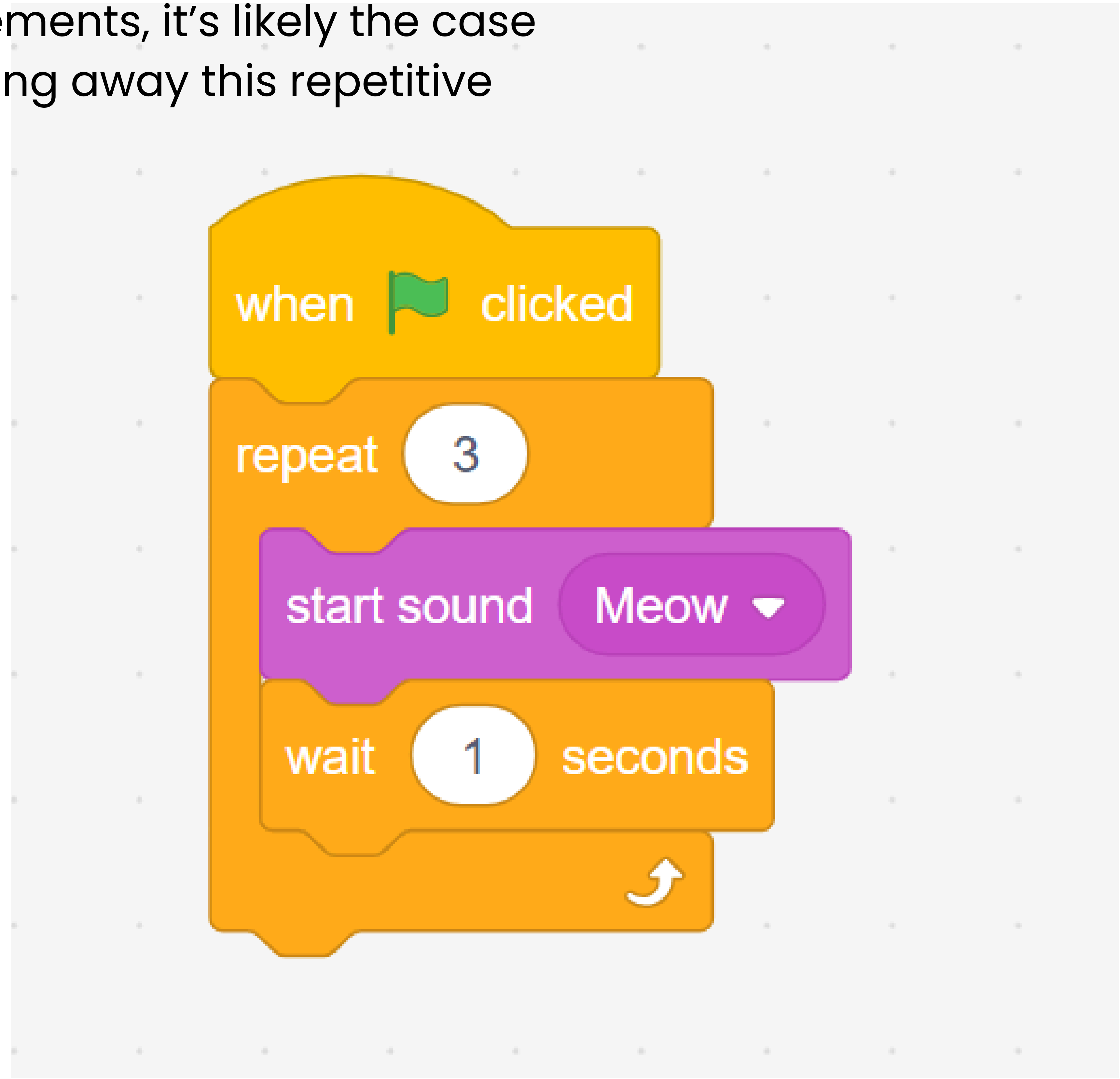

What's your name?


hello , asmaa

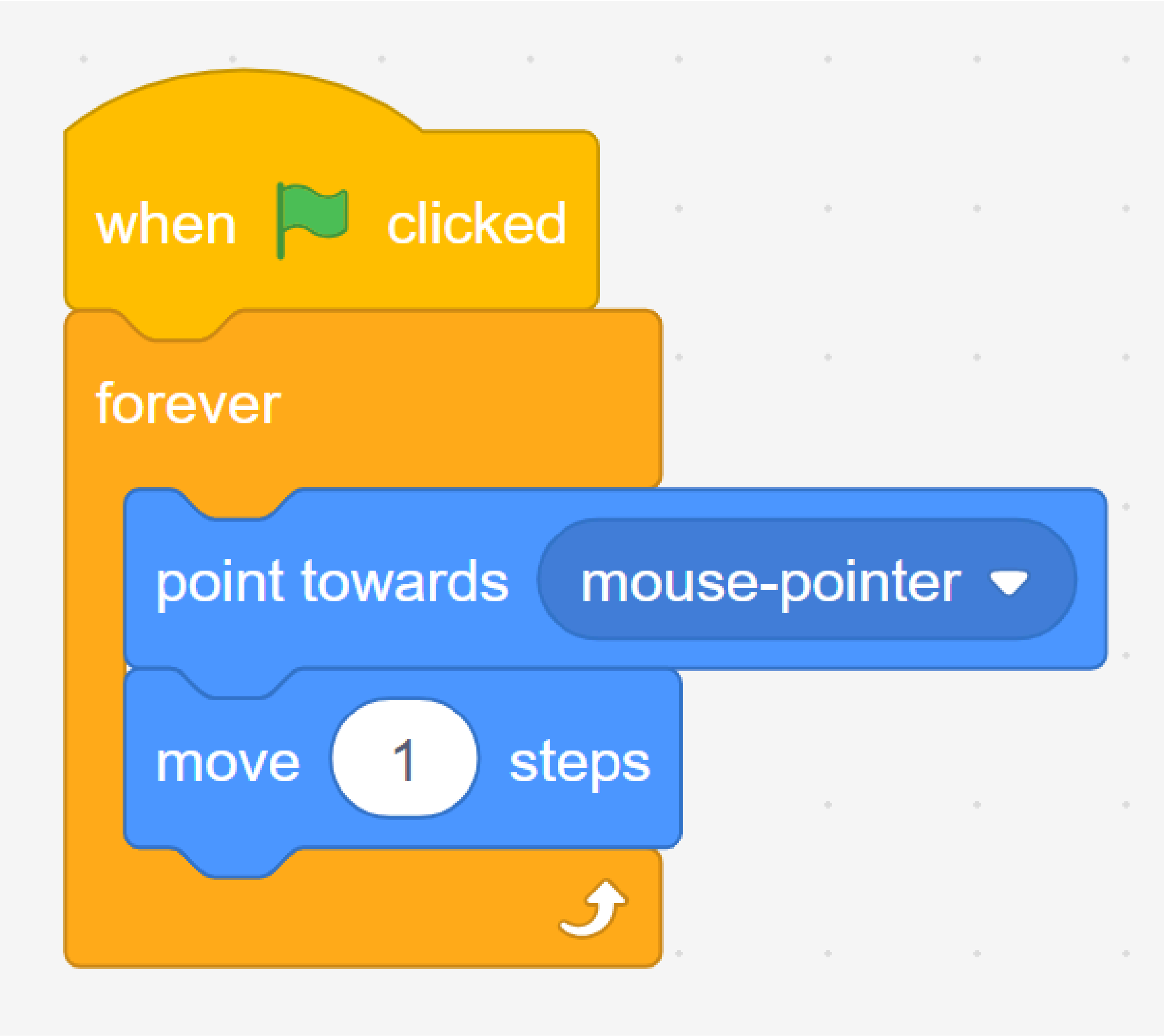
loops

Notice that you are doing the same thing over and over again. Indeed, if you see yourself repeatedly coding the same statements, it's likely the case that you could program more artfully – abstracting away this repetitive code.

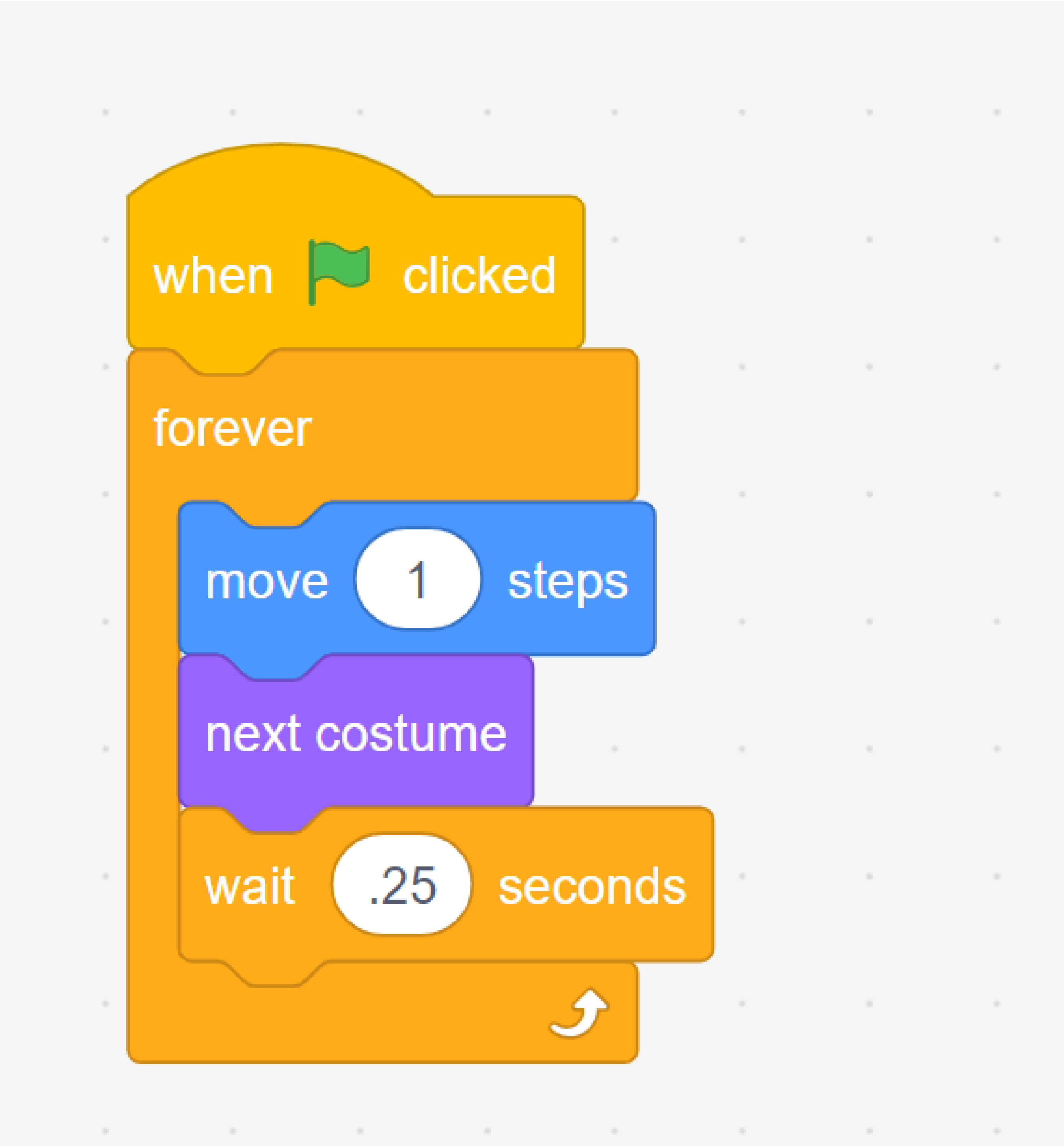


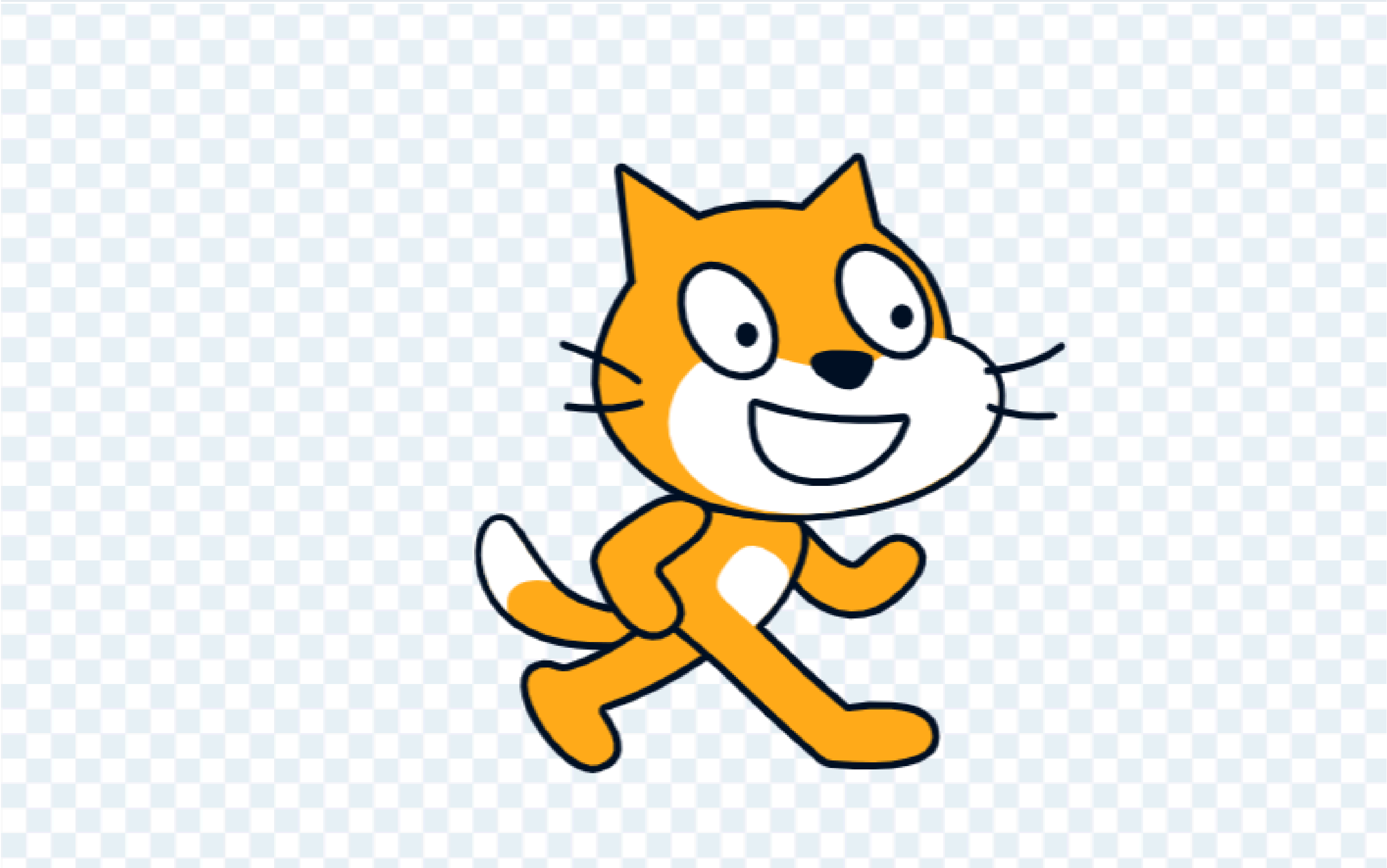


ex2:- when click on mouse move one step point towards mouse .

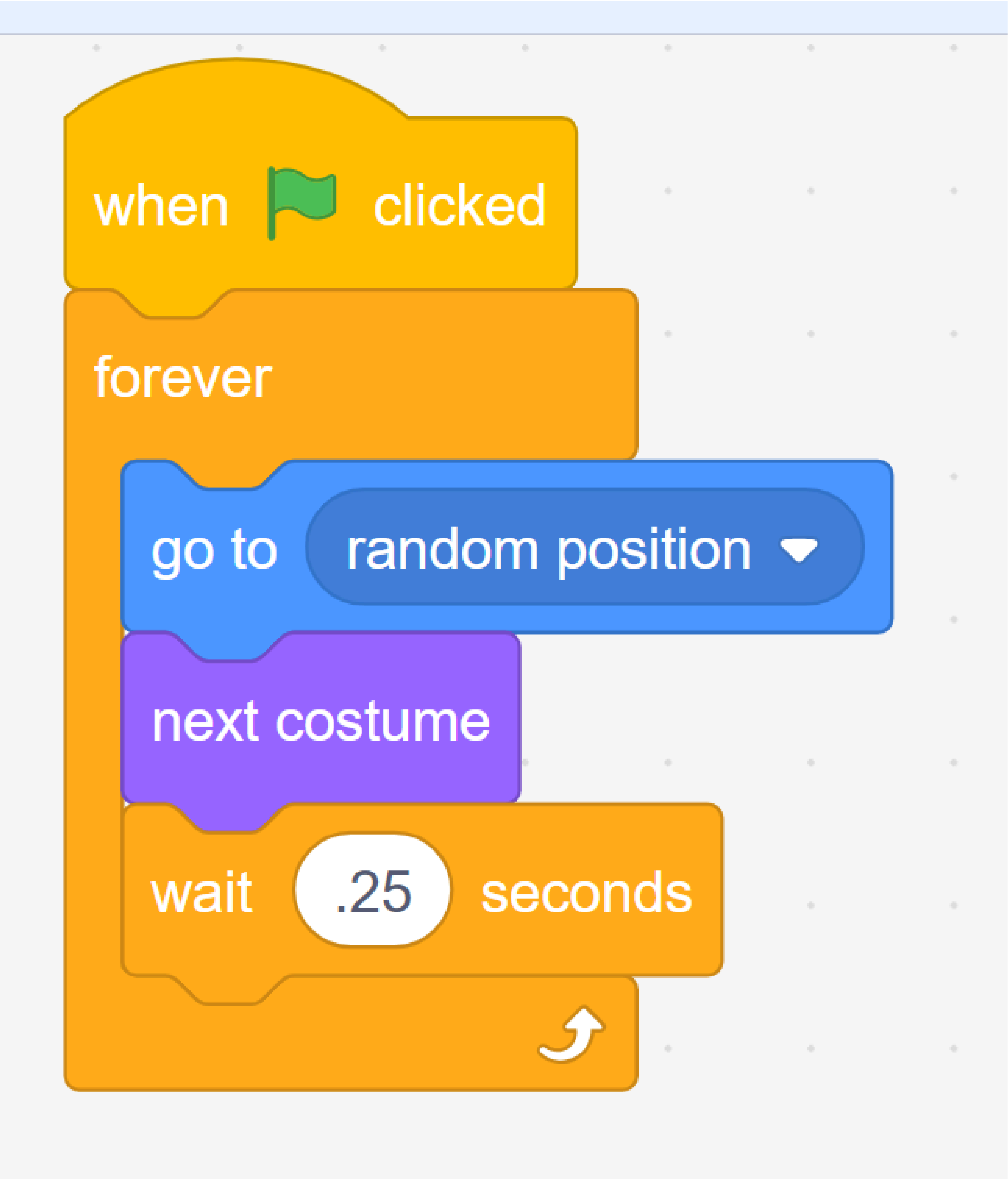


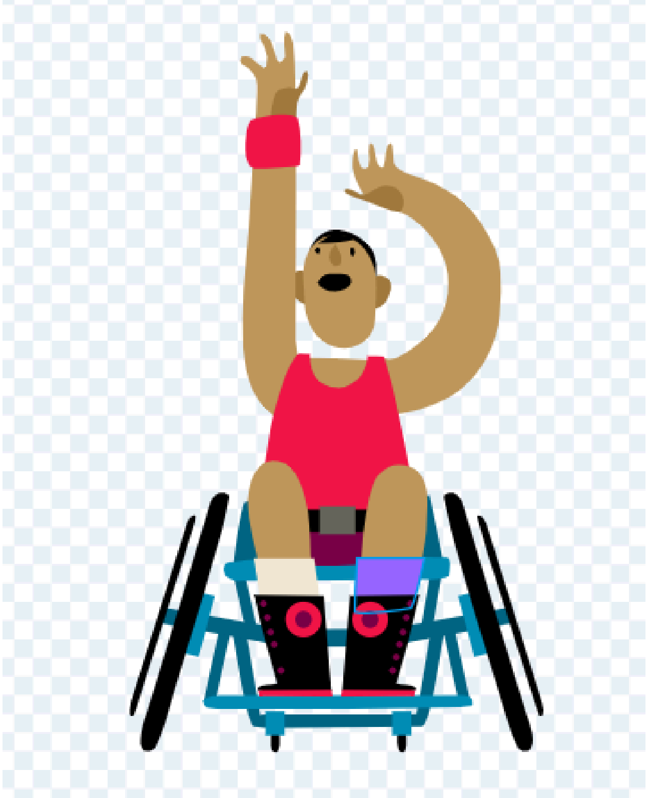
ex3:-

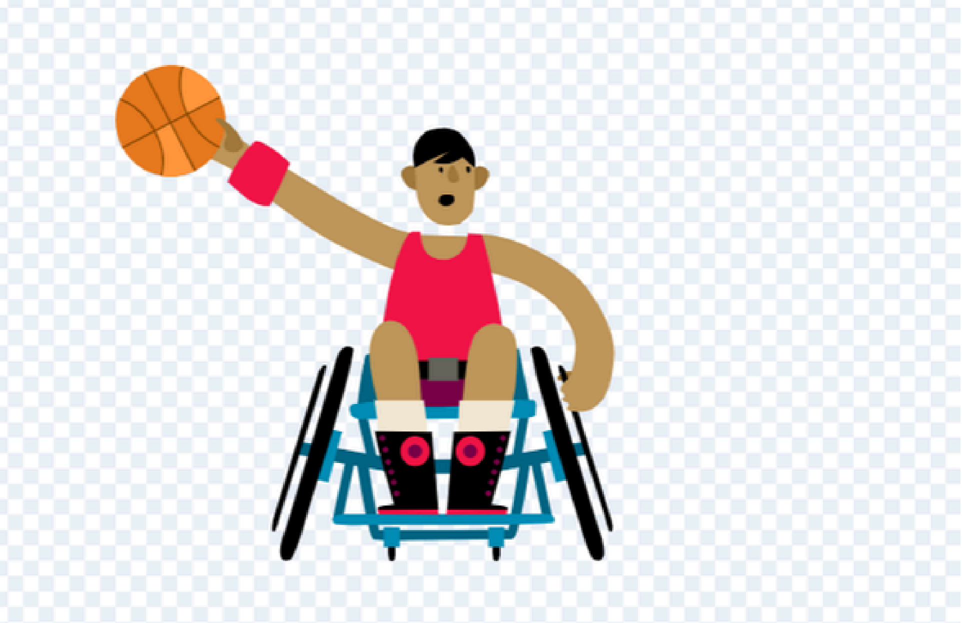

when clicked
forever
move 1 steps
next costume
wait .25 seconds

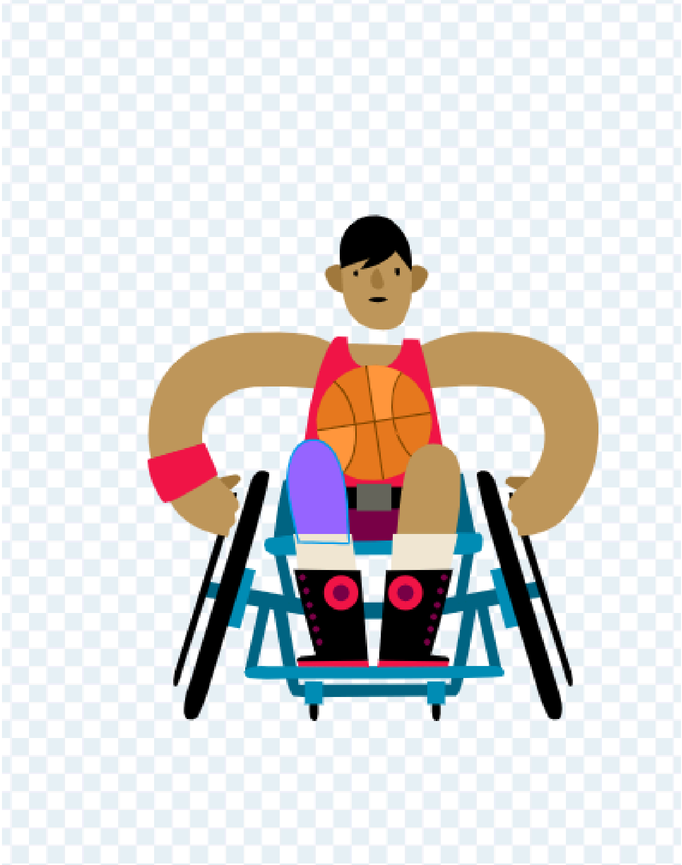


ex4:-

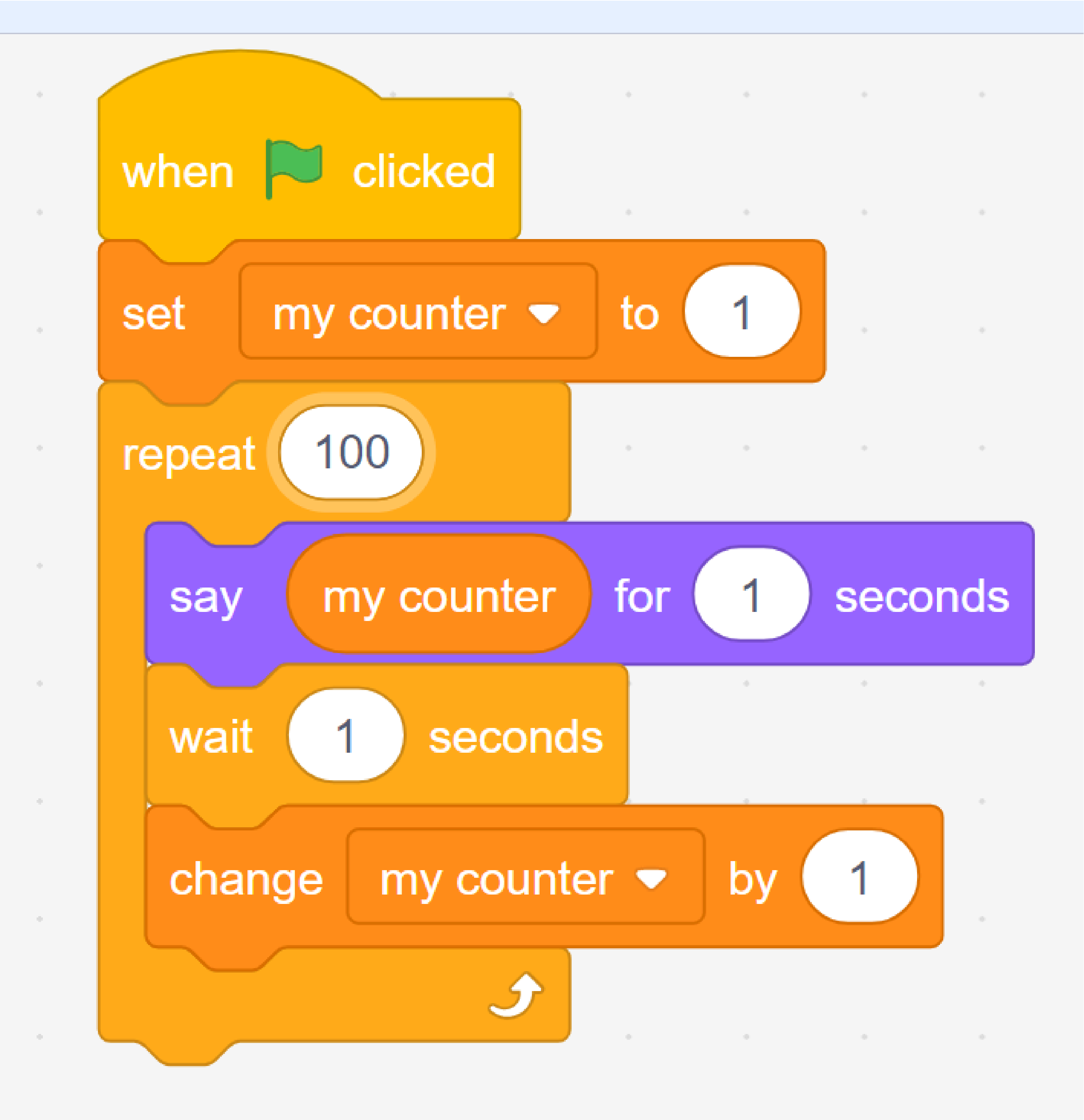

when clicked
forever
go to random position
next costume
wait .25 seconds







ex 5:- count numbers from 1 to 100

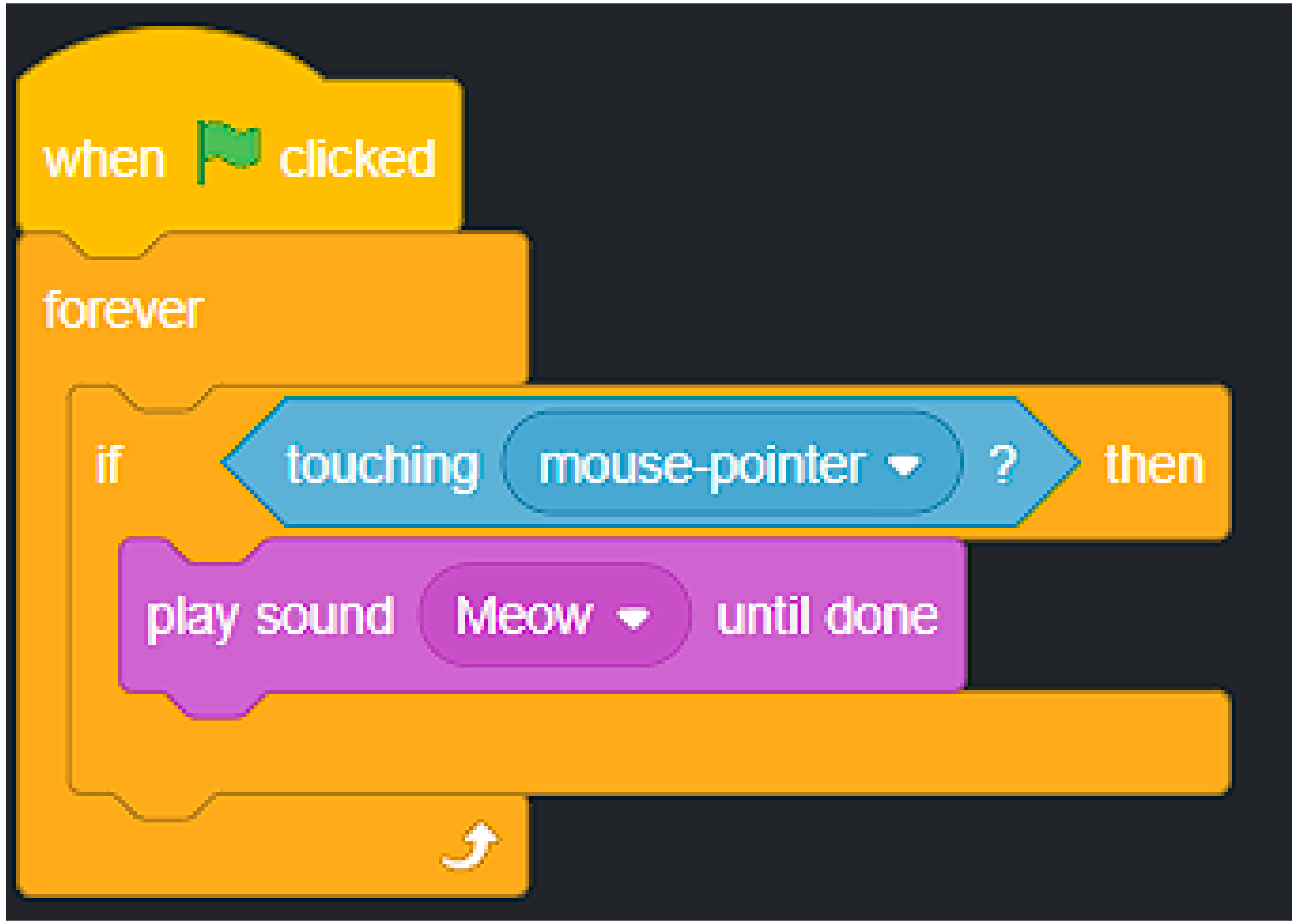


// To do

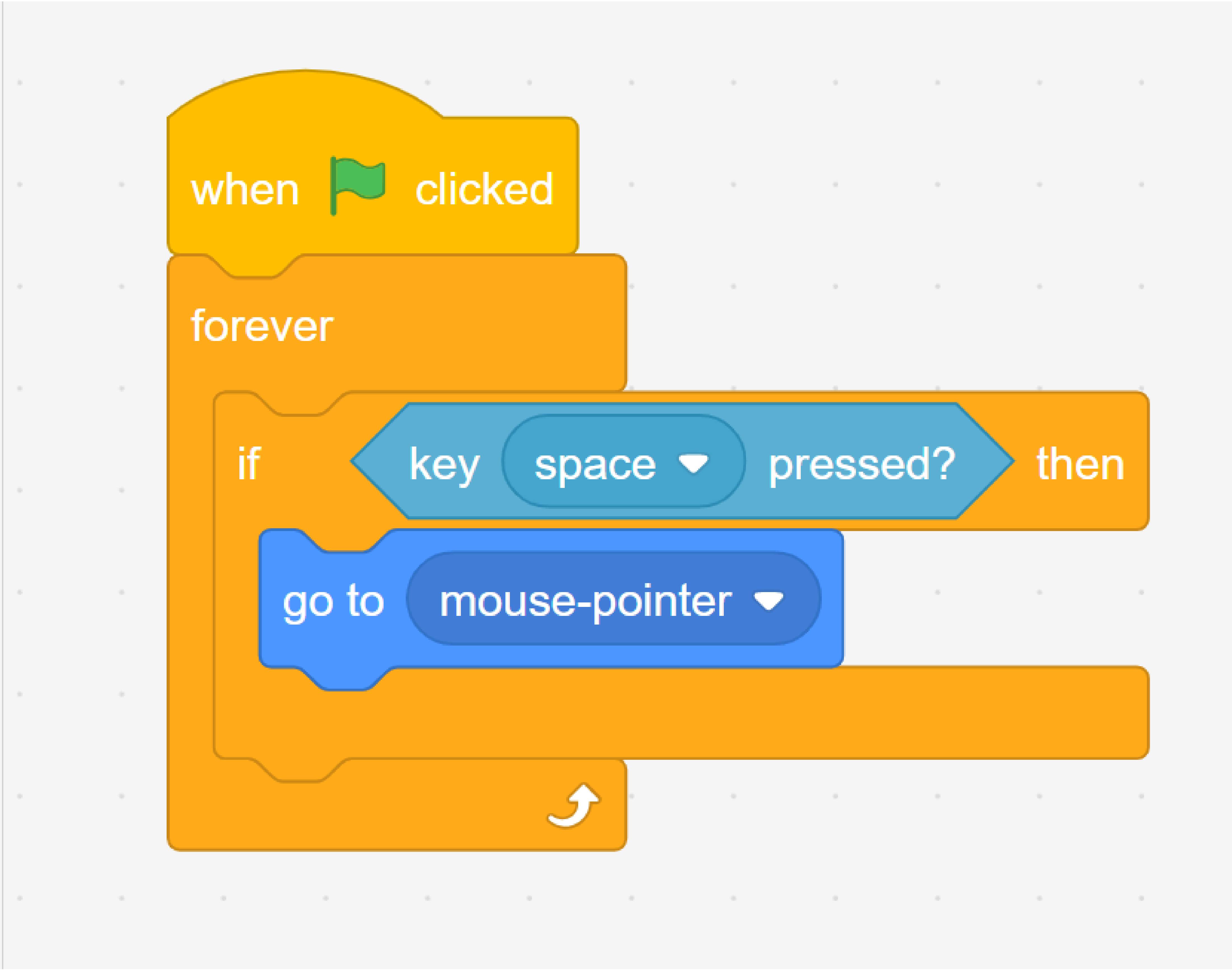
1- make the sprite count odd numbers from 1 to 10

Conditionals

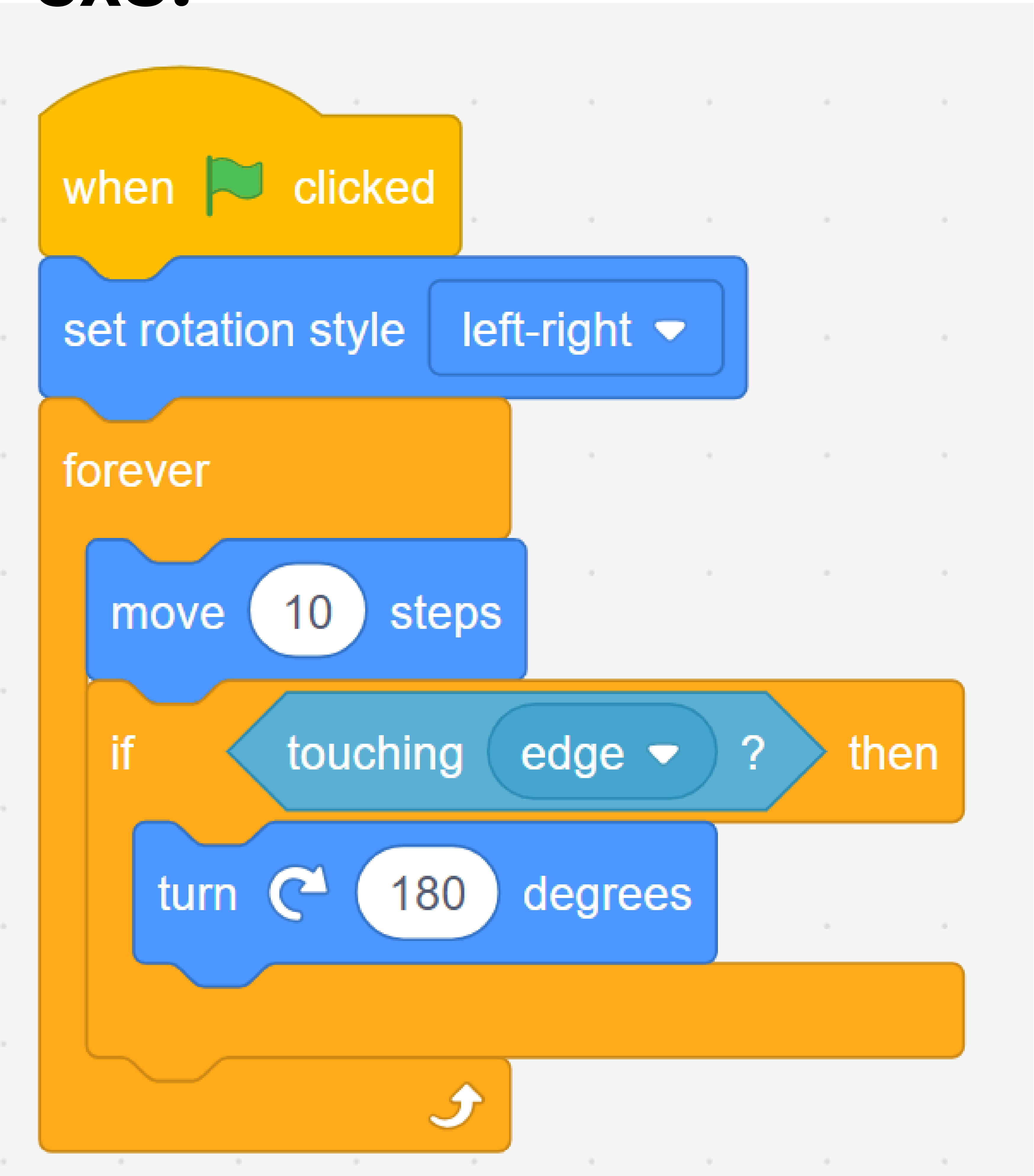
- Conditionals in programming are like road signs that tell the program what to do based on certain situations. If something is true, the program takes one path; if it's not true, it takes another.



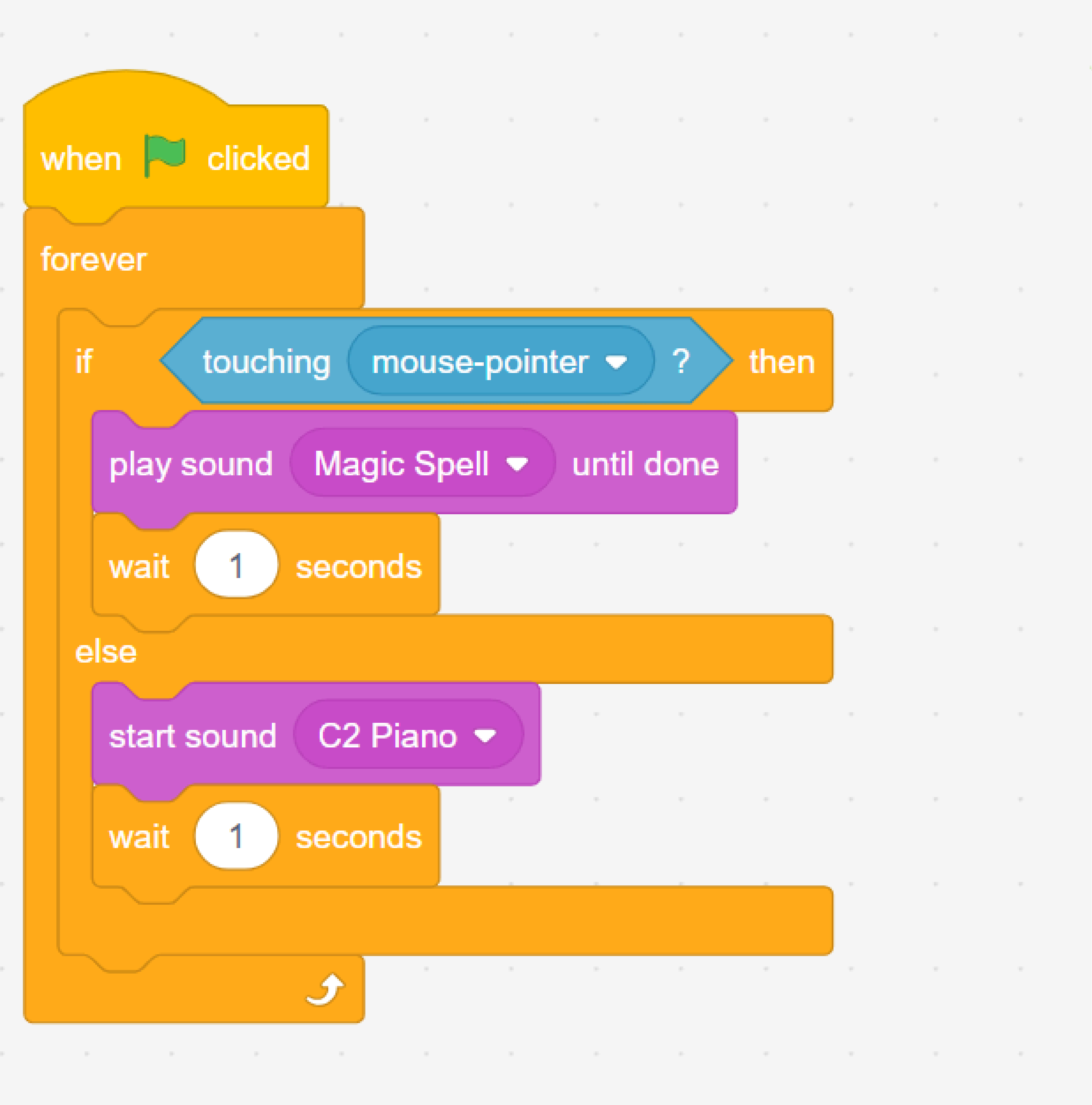
ex 2:-


when clicked
forever
if key space pressed? then
go to mouse-pointer

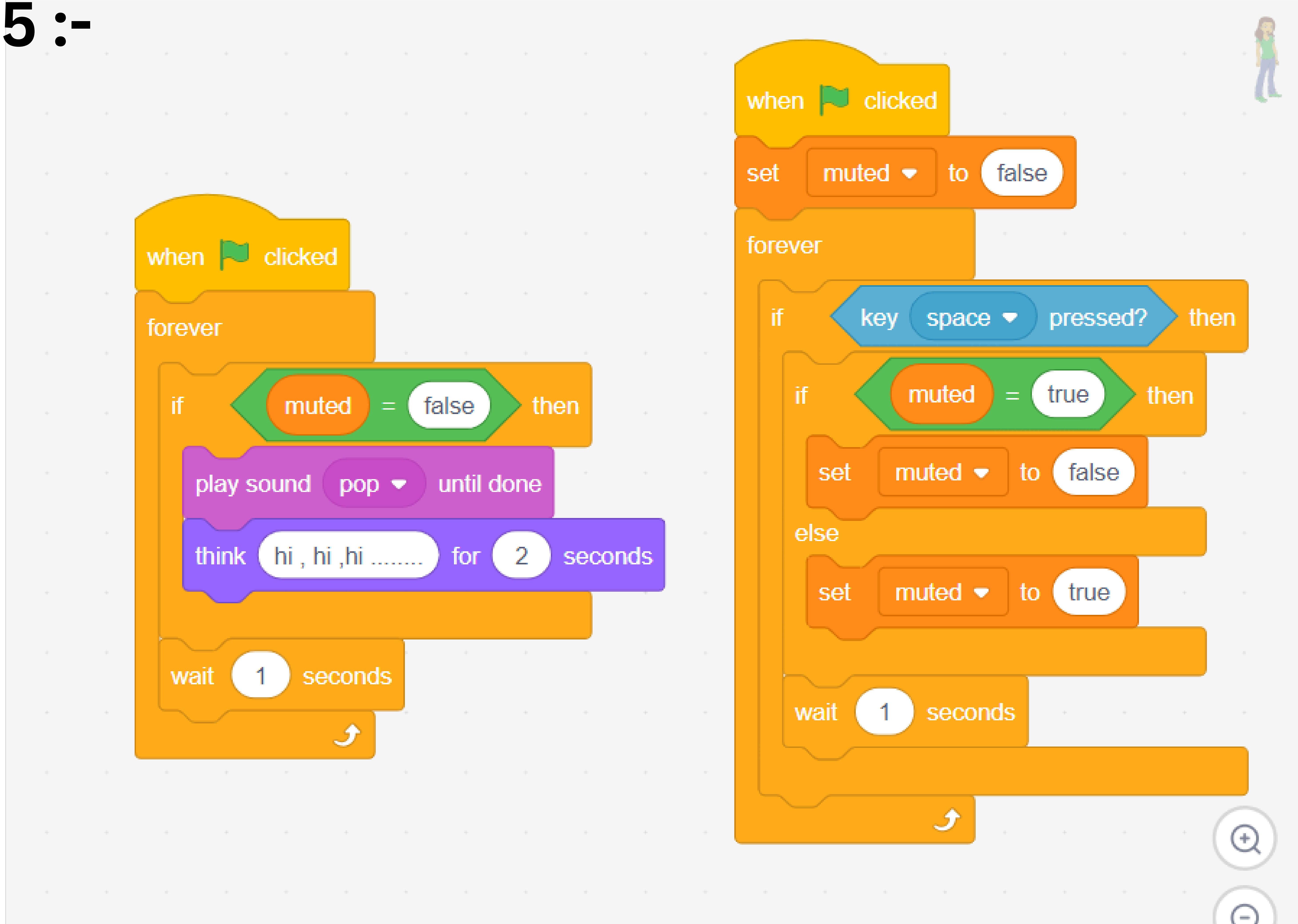
ex3:-


when clicked
set rotation style left-right
forever
move 10 steps
if touching edge ? then
turn 180 degrees

ex4:-


when clicked
forever
if touching mouse-pointer ? then
play sound Magic Spell until done
wait 1 seconds
else
start sound C2 Piano
wait 1 seconds

ex 5 :-

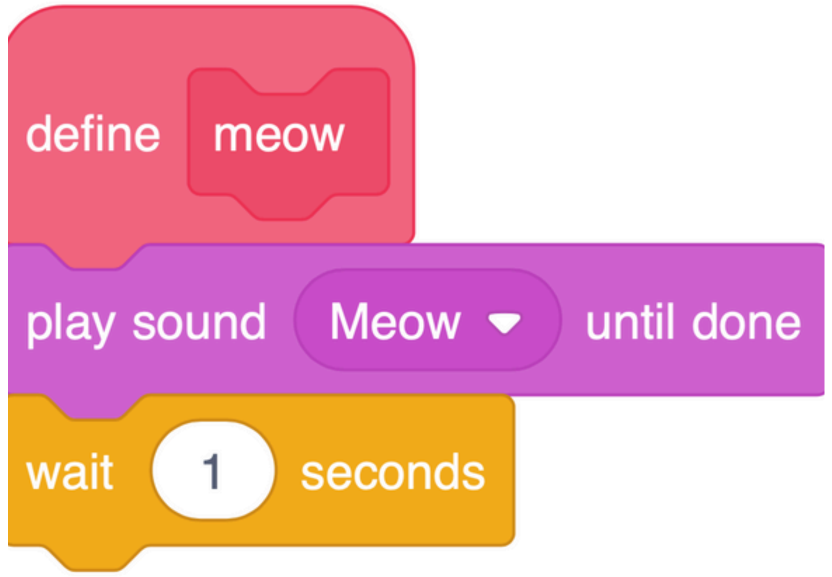

when clicked
forever
if muted = false then
play sound pop until done
think hi , hi ,hi .... for 2 seconds
wait 1 seconds
when clicked
set muted to false
forever
if key space pressed? then
if muted = true then
set muted to false
else
set muted to true
wait 1 seconds

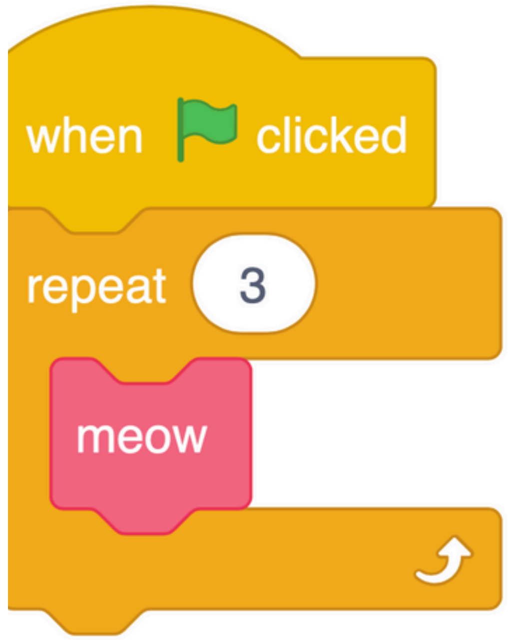
// To do

2- Make the sprite produce a sound when it touches the edge .

Functions

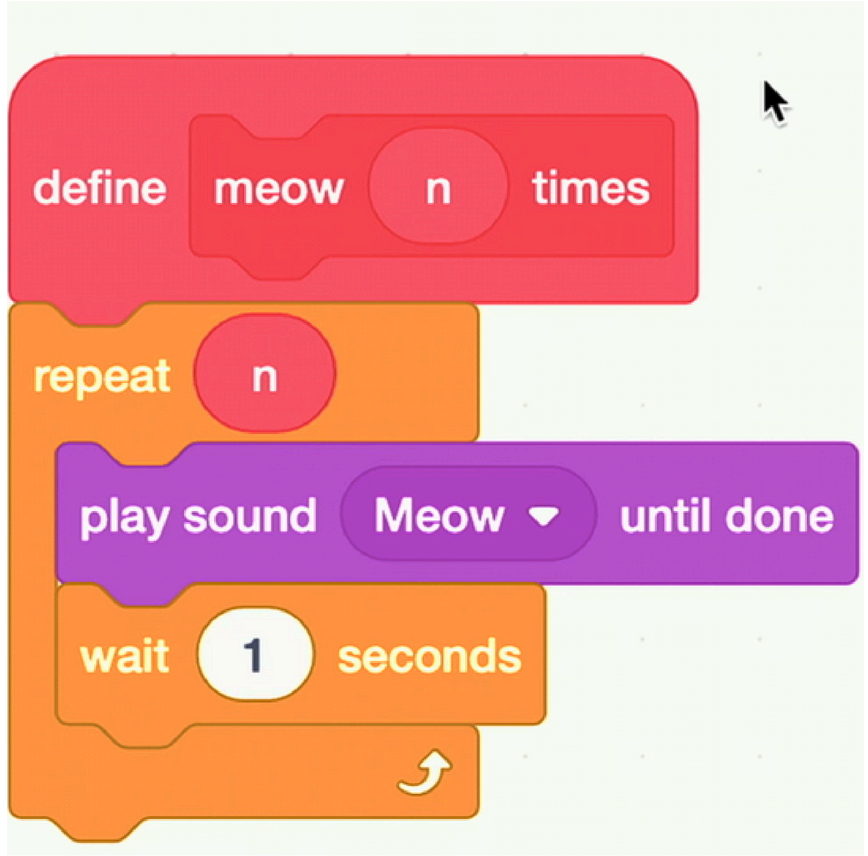
We can even advance this further by using the define block, where you can create your own block (your own function)! Write code as follows:

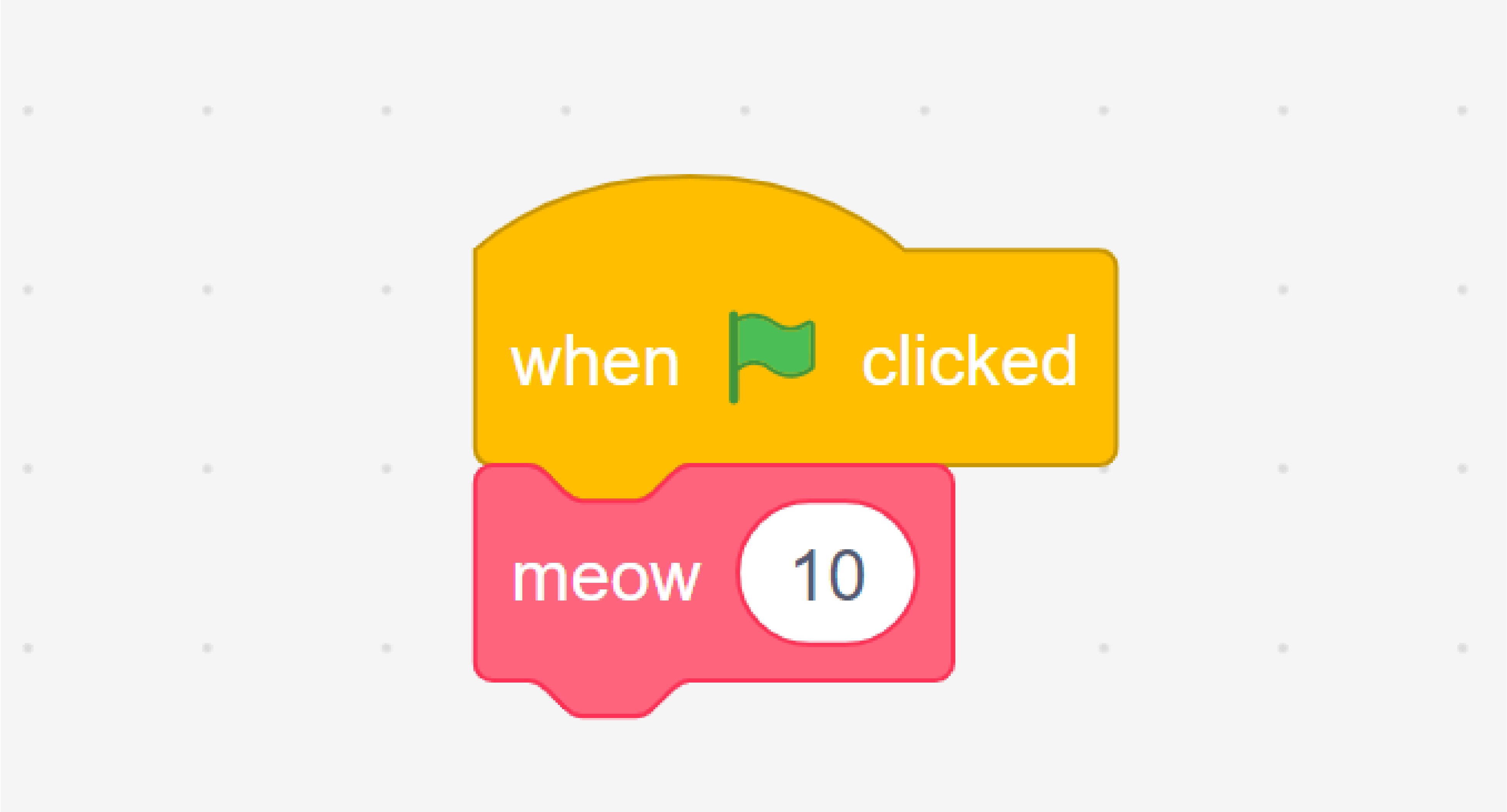




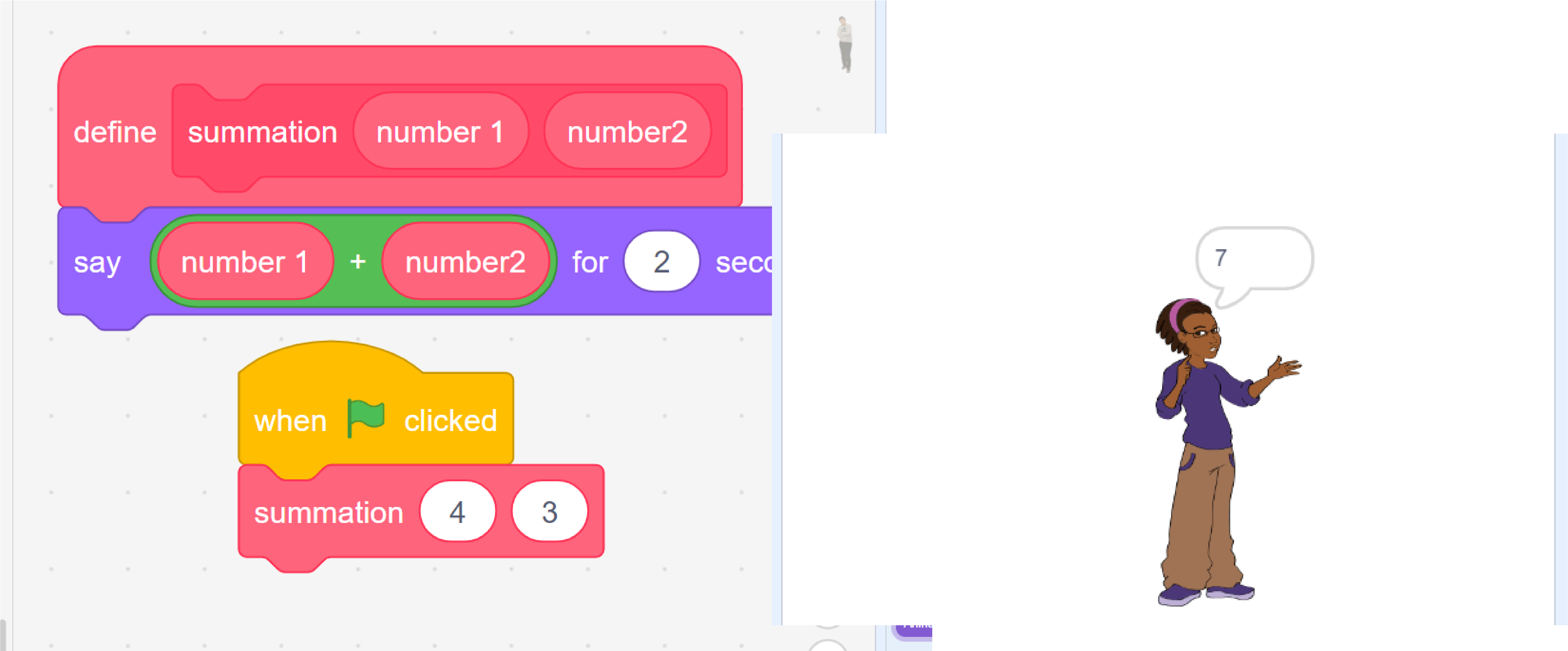
We can even provide a way by which the function can take an input " n " and repeat a number of times:

ex2 :-

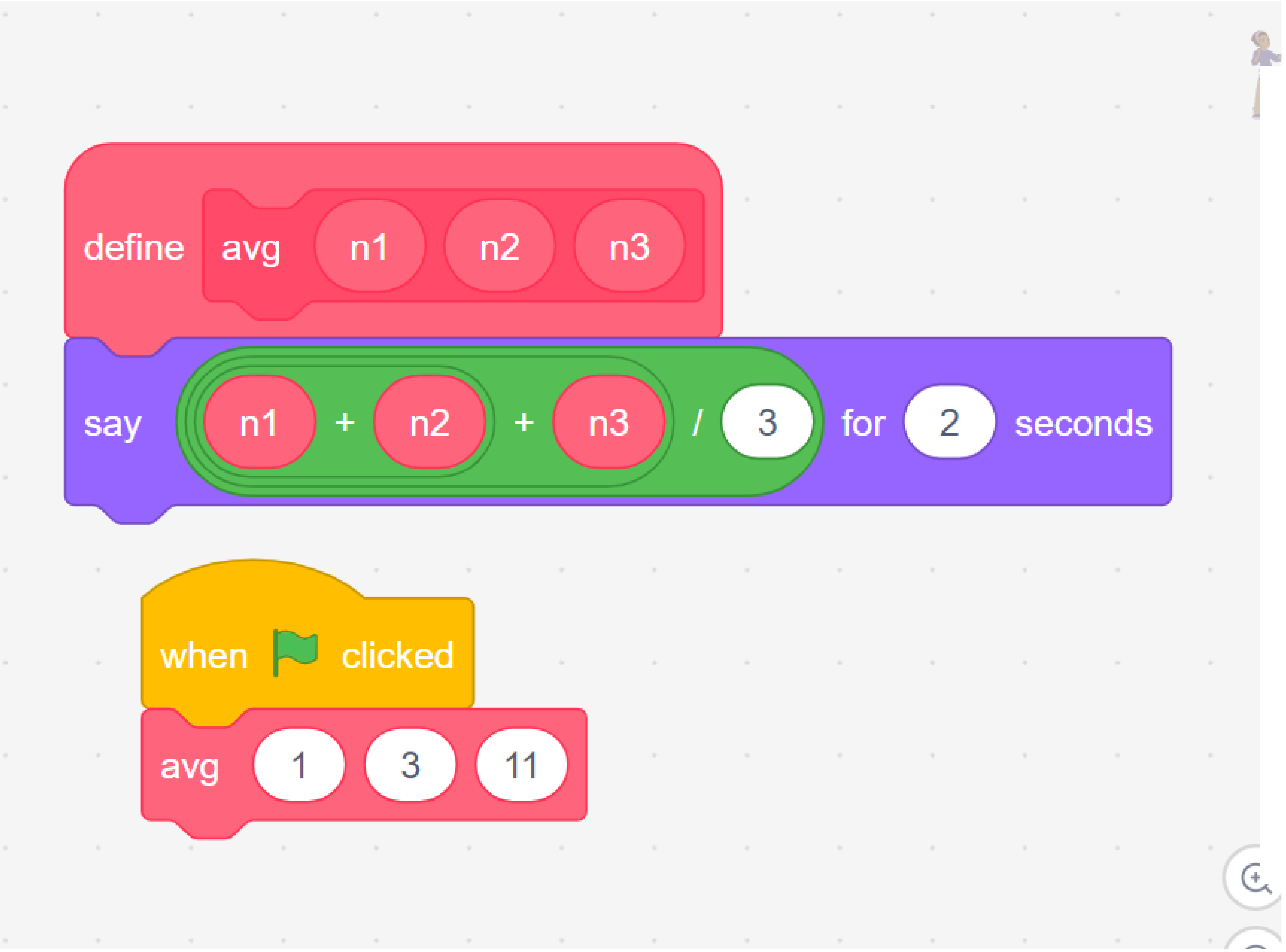


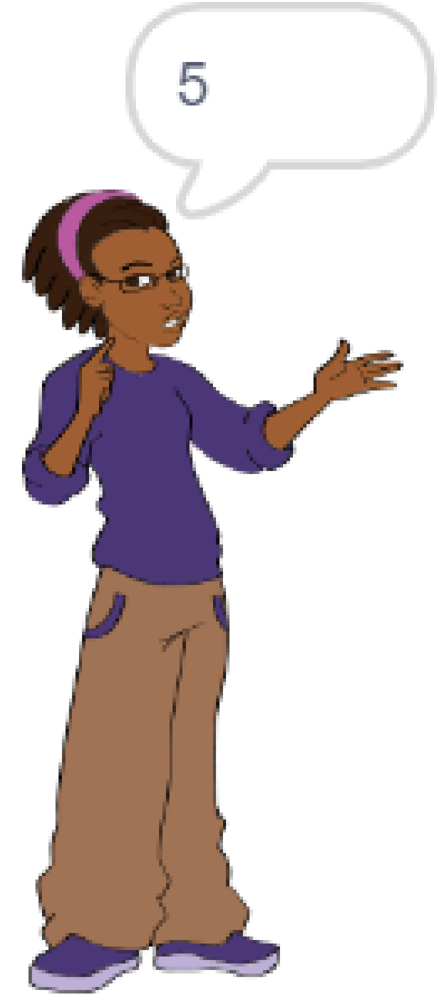


ex 3:-

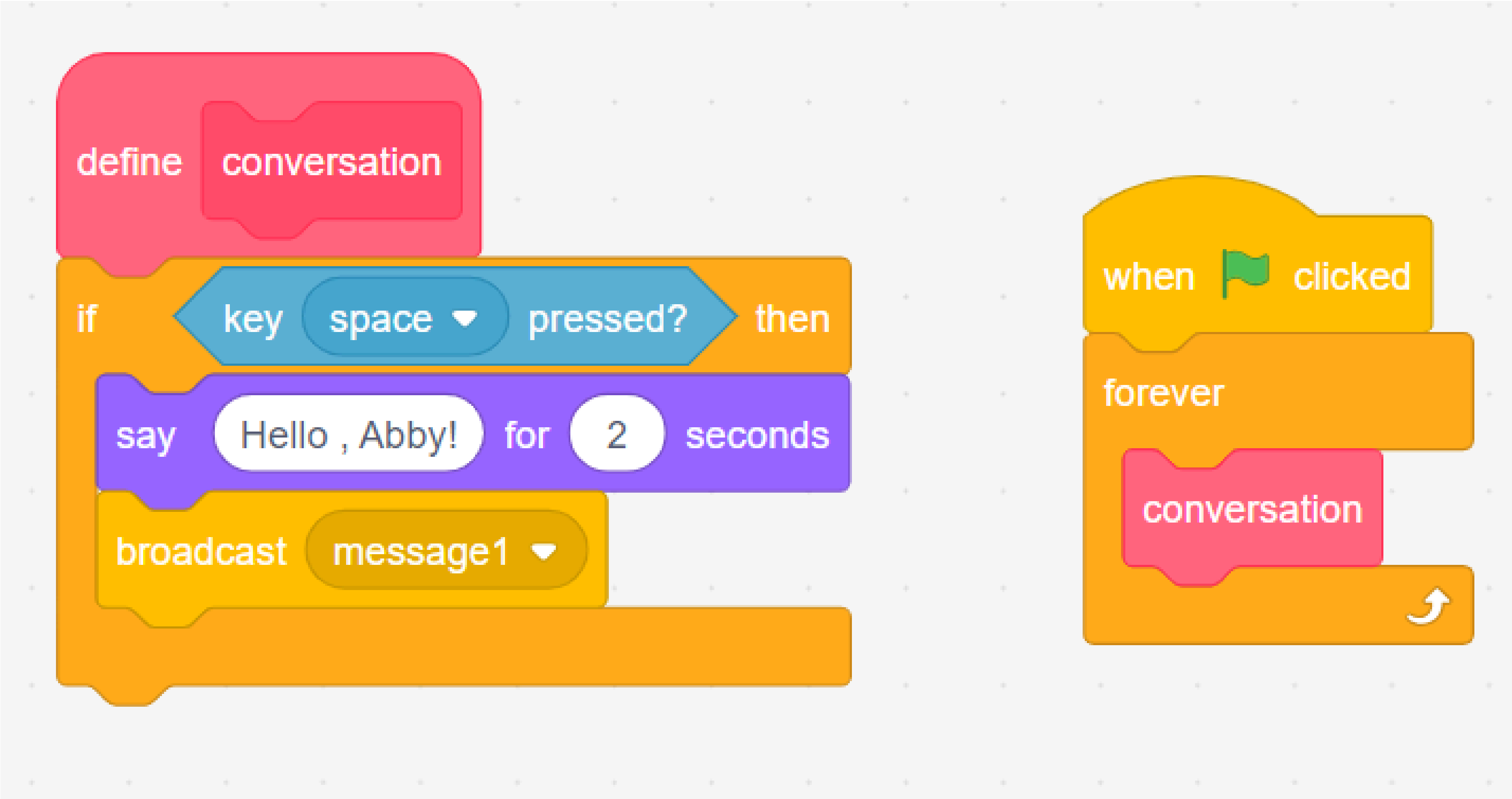

define summation number 1 number2
say number 1 + number2 for 2 seco
when clicked
summation 4 3
7

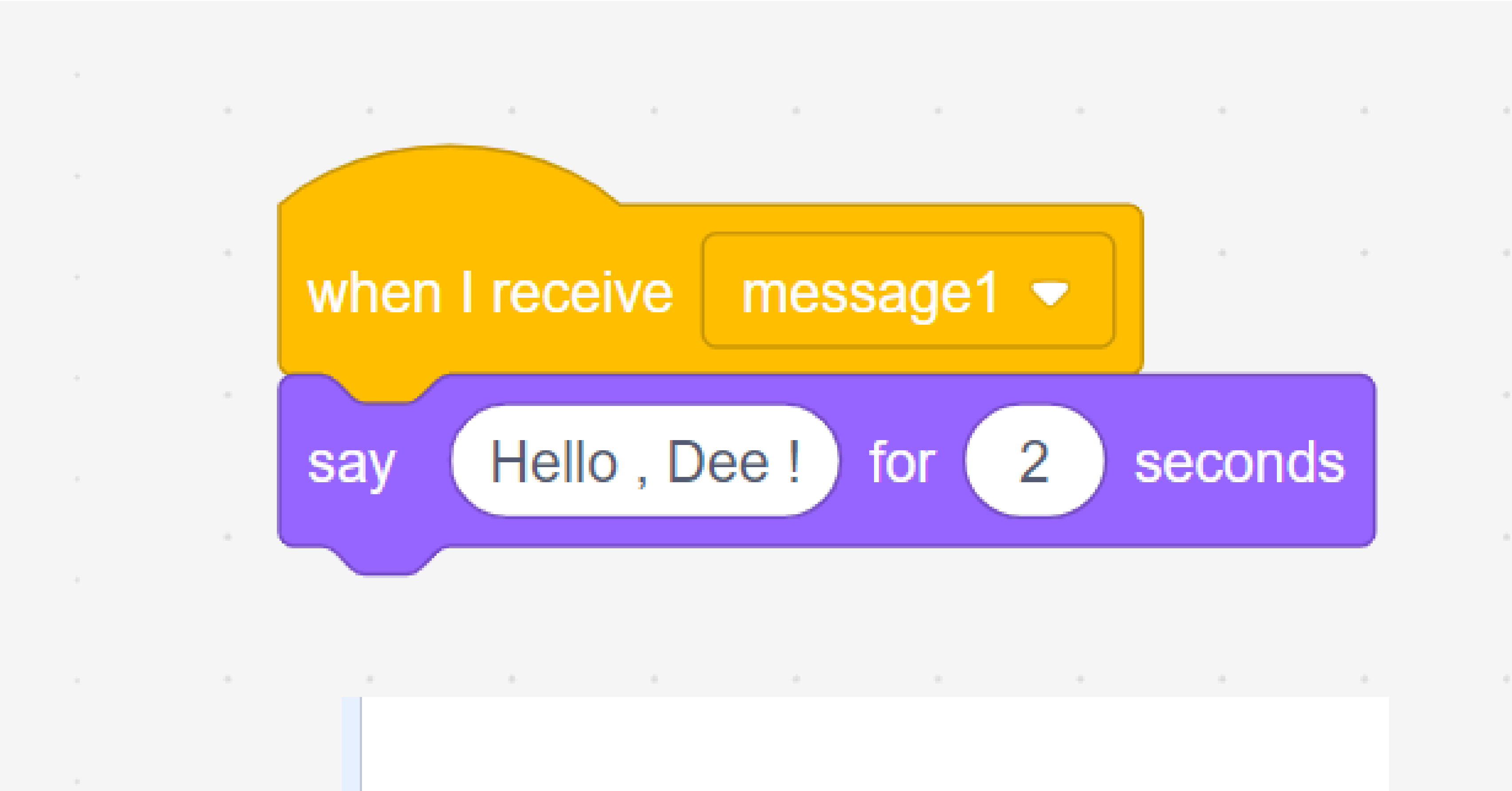
ex 4:- print the average of three numbers

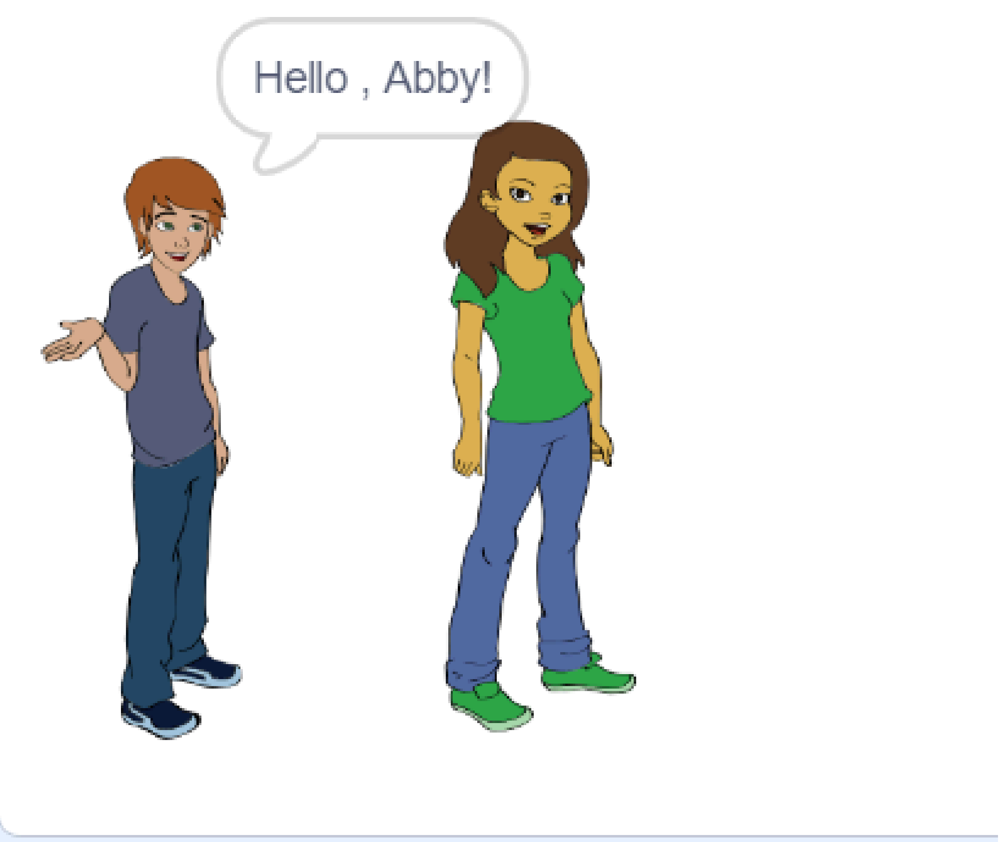


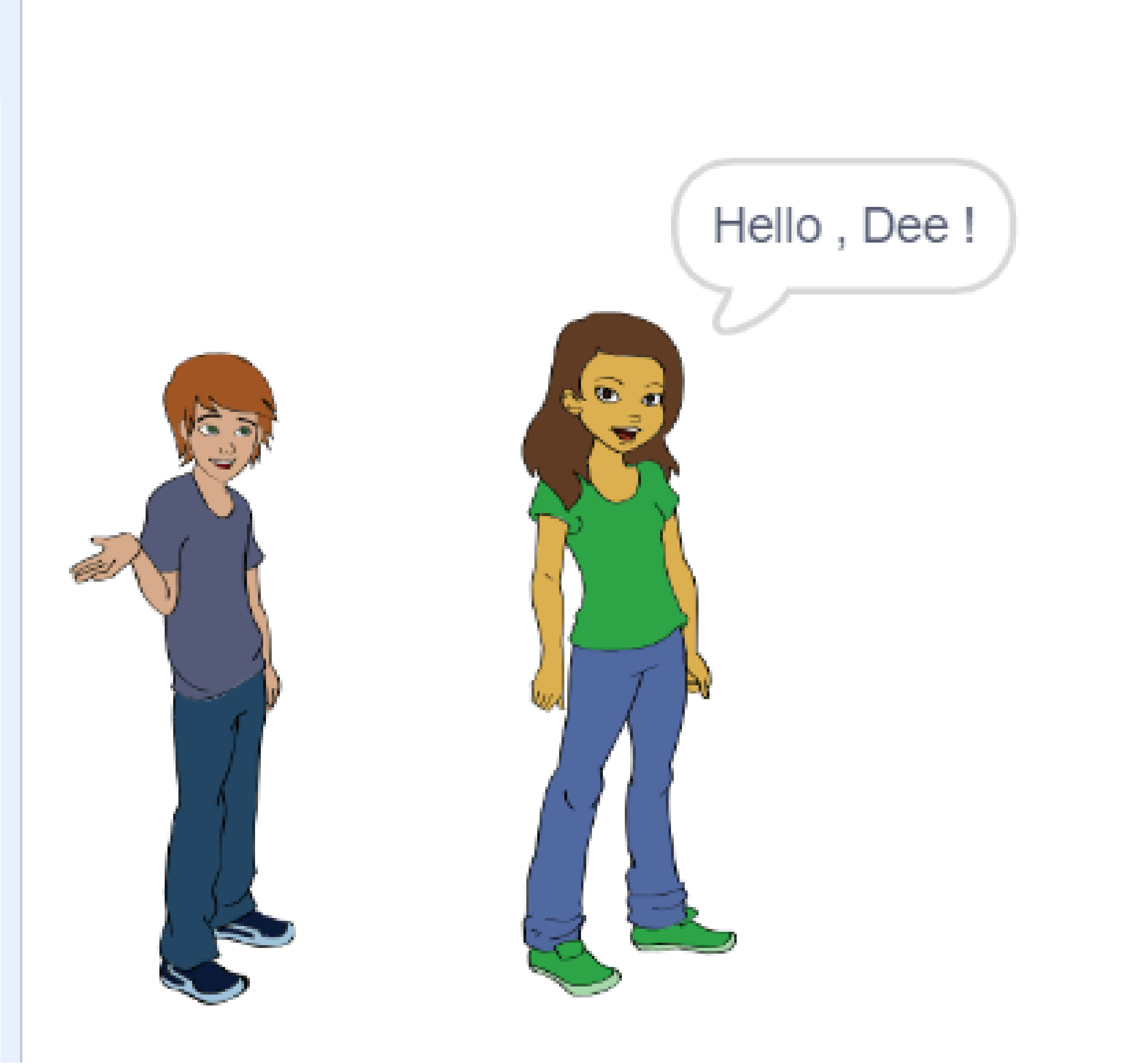


ex 5:-


define conversation
if key space pressed? then
say Hello , Abby! for 2 seconds
broadcast message1
when clicked
forever
conversation


when I receive message1
say Hello , Dee ! for 2 seconds


Hello , Abby!


Hello , Dee !

// To do

3- custom a block take your age as an input and print your age in days (consider the whole year has 365 days)

age * 365